

JUNIOR CERTIFICATE

BIOLOGY

FORM 2

Study Notes

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TOPIC 1: INVESTIGATIVE SKILLS

- An investigation is any activity carried out to discover more about something.
- The methods used in an investigation are called scientific methods.

Planning an Investigation

- A plan is a written account of what one intends to do.

The following should be included in a plan:

- The aim of the experiment
- The hypothesis
- Variables
- Design of the investigation

The aim of the investigation

- The aim is the reason for carrying out an investigation.
- The aim is obtained from the problem that one wants to solve.
- Examples of aims in an investigation include;
- To find out if light is necessary for photosynthesis, to find out the effect of temperature on enzyme activity, etc.

Formulating Hypothesis

- A hypothesis which is a guessed answer is formulated.
- It is a possible explanation of the observation. A hypothesis can be proved to be true or false by results of an investigation.

Identifying variables

- Variables are the things that you measure or can change during an investigation.
- There are three types of variables:

i. Independent or input variable

- This is a variable that is changed during an investigation.
- When an input variable is changed it causes a change in the observations to be made.

ii. Dependent or outcome variable

- This is the variable that you measure or observe during an experiment.
- The dependent variable changes when the independent variable is changed.

iii. Control variable

- This is a variable that is kept constant or is not changed during an experiment. It is kept constant to make sure that it does not influence the dependent variable.

.Designing an Investigation

- It involves listing activities to be carried out in order to get the results from the investigation.
- It includes why the investigation is carried out, what to do, what to use, how to do it, what to observe and how the observation will be handled.
- A design of an investigation is made of the following:

The aim of the investigation

Materials to use during the investigation

- iii. Procedure to use.
- iv. These are the steps followed or done during an investigation
- v. Observations and results
- vi. Conclusions and deductions. This is where the hypothesis is determined to be true or false.

Carrying out Investigations

- Investigations are always carried out systematically during research.
- Before you start your investigation you need to know exactly what you are going to do,
- What results you are going to record and how you are going to record your results.

Recording Data

The observations we make from the investigations are the results obtained.

These results are recorded in various ways.

These can include:

- Description of the changes noted in words
- Records of measurements or numbers obtained in figures. The recorded data can be presented in various ways which include:
 - Tables
 - Linear graphs
 - Bar graphs/charts
 - Histograms **Tables**

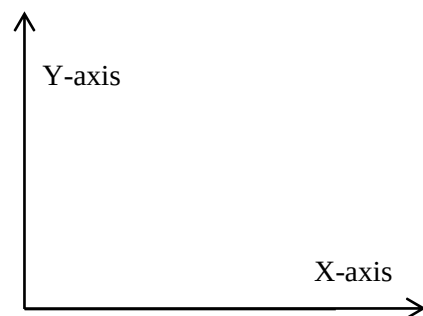
Example

- A farmer had 18 rabbits. He weighed the rabbits and found out that four rabbits weighed 1kg, five rabbits weighed 1.5kg, six rabbits weighed 1.8kg and three rabbits weighed 2.5kg.
- Present the information in form of a table.

Weight of rabbits (kg)	Number
1.0	4
1.5	5
1.8	6
2.5	3

Linear graphs

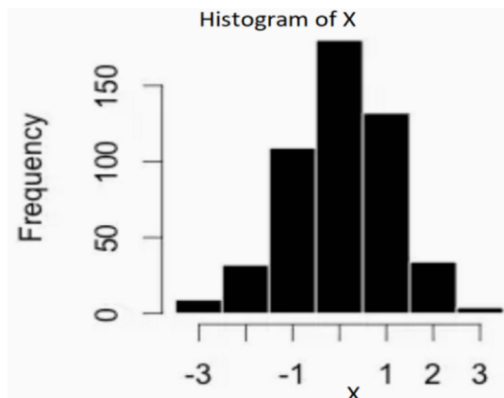
- A linear graph is made of two lines.
- One is vertical and the other is horizontal and are joined at the point of their origin.
- The vertical line is called the y-axis and the horizontal one is called the x-axis.



- The dependent variable is recorded on the y-axis and the independent variable on the x-axis. In between the two lines, points are made where the value of the independent variable meets the dependent variable.
- Then the points are joined using free hand. Use a suitable scale.

Histograms

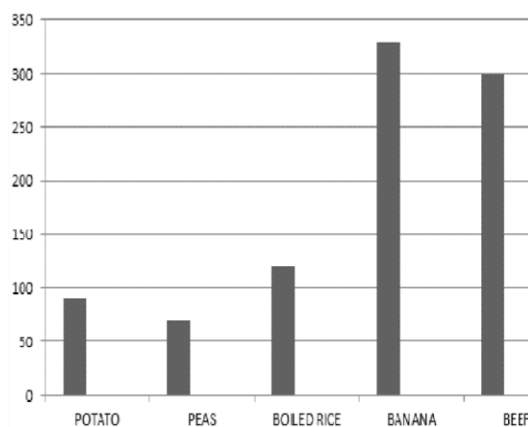
- A histogram is a diagram consisting of rectangles whose area is proportional to the values of a given variable. Its width represents the interval for each set of values. The bars are close together in a histogram.



Example of histogram

Bar graphs/charts

- A bar chart is a graph consisting of rectangular bars whose lengths are proportional to



frequencies in data distribution

- Example of a bar chart

Analysing Data

- Meaning of the data should be explained in relation to the purpose of the investigation.
- This is a careful study of presented data obtained from an investigation.
- Data analysis involves:

- Determining the mean, median, frequency, range
- Looking quantitative relationships between variables which include ratio, direct proportion and inverse proportion.
- Looking for qualitative relationship between variables presented in words or statements instead of figures. This includes correlation, descriptions and explanations of observations made in an investigation.

Drawing conclusions

- A conclusion is either a confirmation or rejection of a hypothesis.
- A conclusion shows whether the aim of an experiment was met or not.
- A conclusion can be presented in form of an opinion, statement, assumption or assertion.

NB: An opinion is a personal statement expressing someone's feeling about an event or situation.

A fact is a statement that is true and can be proved.

Writing reports

- A report is a written record on all that was done during the investigation.
 - The report should have the following information:
- A title taken from the aim of the experiment
 - Statement of the aim of the experiment
 - Statement of hypothesis
 - Description of the procedure used
 - Results obtained
 - Analysis of the results (can be presented in forms of tables and graphs)
 - Conclusion made from the data

Revision Exercise

1. a. What is a variable?
b. Mention any two types of variables
c. State any two items that can be included when planning an investigation.
2. The table below shows the effect of amount of fertilizer applied on maize yield per hectare.

Amount of fertilizer (kg)	Maize yield (kg)
50kg	1,000
150	2,500
250	3,500
300	3,500

- a. From the table identify the independent and dependent variable
- b. Suggest two possible control variables in the experiment
- c. What is the correlation between the dependent and independent variable
3. Write a report to your biology teacher on increased cases of corona virus at the school. In your report include the aim of the report, the methods you used to collect data, the results of your investigation on the increase in cases and conclusion.
4. a. Design an investigation that you can carry out to show that moisture affects growth of microorganisms.

TOPIC 2: PLANT STRUCTURE AND FUNCTION

- Flowering plants are divided into **three** main parts.

1. stems

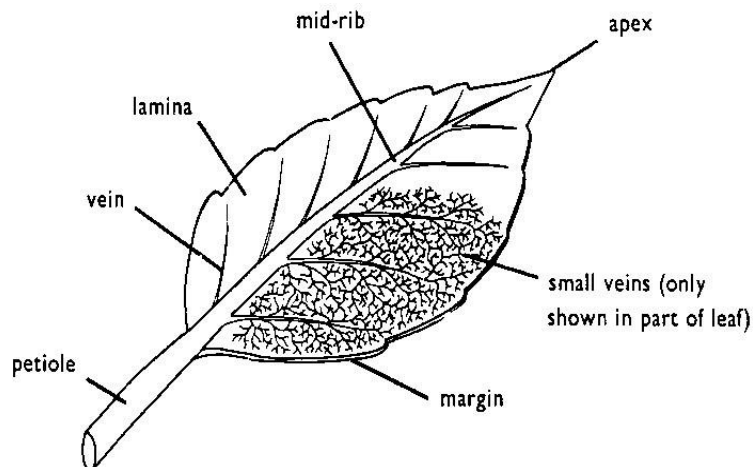
2. roots

3. leaves

Functions of parts of a plant

a. leaves

- They make food for the plant during photosynthesis
- They are a site for gaseous exchange
- They act as food storage organs
- They give support to the upper parts of the plant.
- They act as storage organs
- Some are used for vegetative
- It also contains vascular bundles



FUNCTIONS OF EXTERNAL PARTS OF THE LEAF

Lamina

- where most chloroplasts are located
- It is flat and thin for exposure to sunlight and oxygen

Petiole

- Attaches the leaf to the stem there by exposing it to sunlight for photosynthesis
- It contains vascular bundles, xylem and phloem
- Xylem transport water to the leaf while phloem transport food away from the leaf

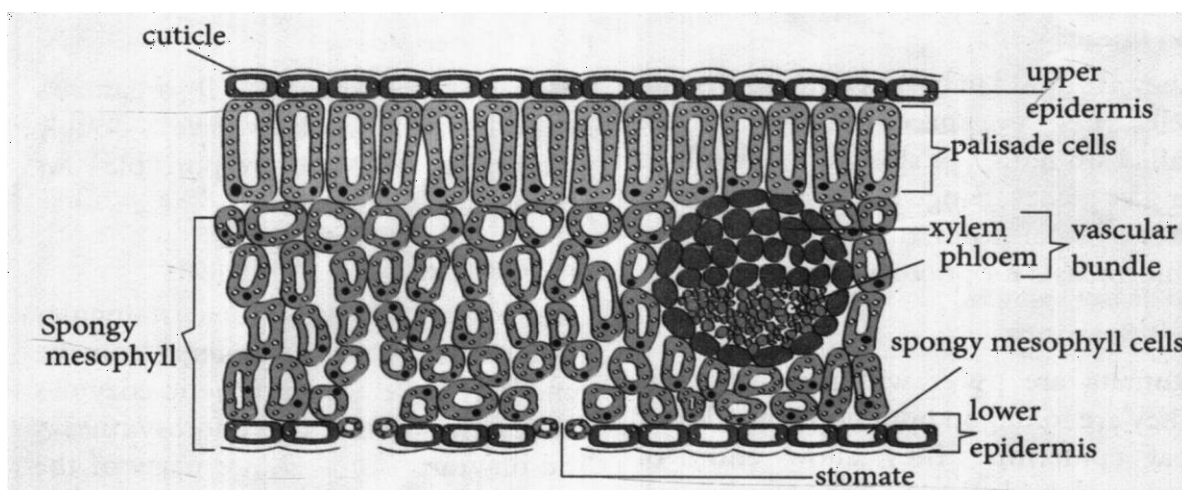
Mid rib

- Holds the lamina in position.
- Have vascular bundles for transport of substances to and from the leaf.

Cuticle

- It is water proof which prevents excessive water loss from the leaf surface by evaporation
- Protects internal parts of the leaf from physical damage
- Hairy cuticle protects the leaf from predators

FUNCTIONS OF INTERNAL PARTS OF THE LEAF



Epidermis (both upper and lower epidermis)

- Consists of one layer of thin flattened cell sthat have no chloroplasts
- Outer walls are covered with cuticle a waxy and water proof substance produced by epidermal cells
- Upper and lower epidermises contain small pores called stomata surrounded by pairs of guard cells

The epidermis plays the following functions

- It protects the inner parts of the leaf from damage

- It contains stomata which are entry and exit for air into and out of the leaf.
- Its cells allow light to penetrate since they have no chlorophyll

Palisade mesophyll

- This is a layer of cells located below the upper epidermis
- The cells are elongated and column shaped
- Palisade mesophyll cells contain a lot of chloroplasts hence is the main site of photosynthesis
- Palisade cells are closely packed with no air spaces

Function of palisade cells

- It is the main site for photosynthesis

Spongy mesophyll

- A layer of cells located between the palisade mesophyll and lower epidermis
- The cells are irregular shaped which are fit loosely
- They have a lot of air spaces between them which allow air circulation and gases
- The spongy mesophyll cells are lined with moisture to aid gaseous exchange

Functions of spongy mesophyll cells

- It is a site of photosynthesis but has fewer chloroplasts than palisade cells
- It is a site of gaseous exchange in plants
- The spongy mesophyll cells store starch produced during photosynthesis
- It also stores water to be used in photosynthesis

Vascular Bundles (phloem and xylem)

- Consist of xylems and phloem.

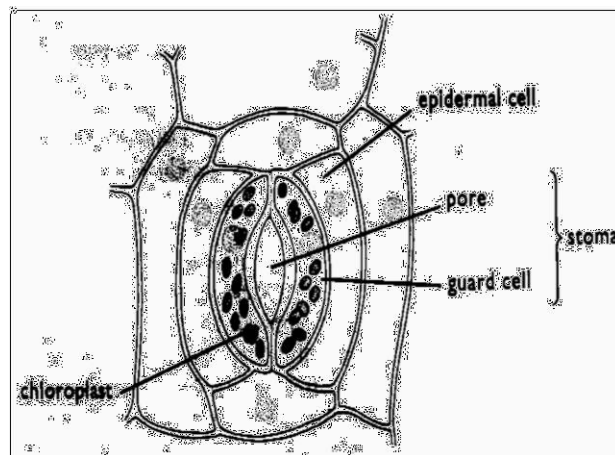
Functions of vascular bundles

Xylem

- Conduct water and mineral salts from the roots to the leaves

Phloem

- Transport food materials from the leaves to other parts of the plant



Stomata

These are pores located on both the upper and lower epidermis

Function of stomata

- Allow gaseous exchange between the leaf and atmosphere
- It allow transpiration to take place

Activity: To investigate Distribution of Stomata in leaves

Materials

- A fresh leaf fresh sisal leaf
- Microscope
- Scalpel or razor blade
- Distilled water

Procedure

1. Using a tip of scalpel peel off a small piece of skin from upper surface of the sisal leaf
2. Pull off a thin skin from the leaf
3. Place the skin on the microscope slide and cover it with a drop of water
4. Place a coverslip on top and mount it on a microscope
5. Examine it using lower objective lens of the microscope
6. Count the number of stomata that are observed
7. Repeat the procedure with a skin from the lower surface of the leaf.
8. Compare the number of stomata on the upper and lower surfaces of the sisal leaf

Results

You observe more stomata on the lower surface than on the upper surface of the sisal leaf.

Conclusion

The lower surface of the sisal leaf has more stomata than the upper surface.

NB:

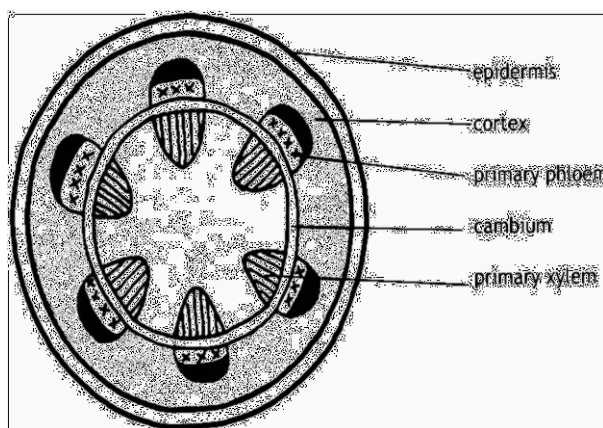
- There are more stomata on the lower surface of the leaf to reduce rates of water loss through stomata due to direct sunlight.
- Plants that grow in areas with abundant water supply have many stomata to help eliminate excess water.
- Plants that grow in areas with inadequate water have few stomata to minimize water loss through transpiration.

Adaptations of the leaf for photosynthesis

- a. The leaf has chloroplasts which contain chlorophyll that absorbs sunlight for photosynthesis
- b. The leaf has thin lamina to allow light to penetrate to all cells and to allow faster diffusion of oxygen and carbon dioxide
- c. It has stomata which allow gases in and out of the leaf
- d. It has branching network of veins which have vascular bundles to transport water to the leaves and glucose to other parts of the plant.
- e. The leaf has air spaces in the spongy mesophyll for exchange of oxygen and carbon dioxide
- f. It has transparent cuticle to allow light to pass through and reach cells for photosynthesis.
- g. It has no chloroplasts in the epidermis this is to allow sunlight to penetrate to mesophyll layers.

THE STEMS

The structure of the stem



Epidermis

- This is the outer single layer of cells in the stem. It protects the inner delicate parts from mechanical damage.

Cortex

- It is made of three types of cells which are
- Parenchyma cells which form the bulk of cortical tissue
- Collenchyma cells that strengthen the tissues
- Sclerenchyma provide mechanical support

Xylem

- It is part of the vascular bundle and it performs the following functions
- Transports water and dissolved mineral salts in the plant
- Provides support to the plant because it contains a tough material called lignin.

Phloem This transports organic food substances in the plant

NB: In monocot stems vascular bundles are scattered while in dicots they are arranged in form of a ring.

ROOTS

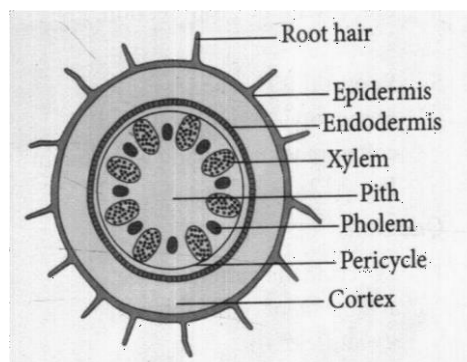
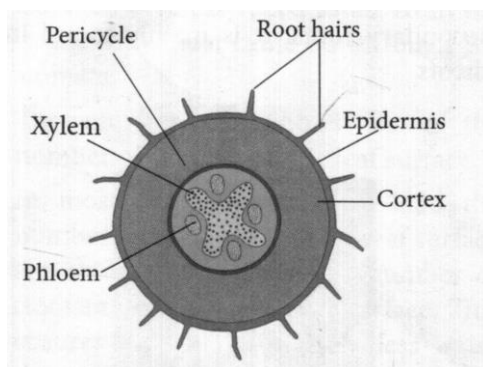
The Cross-section of the root

Epidermis: The outer layer of cells. It protects inner parts of the root.

Root hairs: They are extensions of the epidermal cells.

- They are used to absorb water and mineral salts from the soil.

Cortex: Part of the root located between the epidermis and the endodermis.



- It is used for storage of substances in other plants eg potatoes and cassava

Pericycle: This is a layer of cells immediately after the endodermis.

- Branch roots arise from the pericycle.

Vascular bundles: They transport substance in the plant.

THE PROCESS OF PHOTOSYNTHESIS

- Photosynthesis is the process in which plants make glucose from water and carbon dioxide in the presence of sunlight and chlorophyll.
- Water and carbon dioxide are the raw materials for photosynthesis.
- Glucose is the main product of photosynthesis while oxygen is the by-product.

Word Equation for Photosynthesis

Water + Carbon Dioxide $\longrightarrow$ Glucose + Oxygen

Chemical Equation for Photosynthesis



- Photosynthesis takes place in two stages:

Light stage

- It is the first stage of photosynthesis and depends on light.
- Light energy from the sun is absorbed by chlorophyll
- The light is used to split water into hydrogen and oxygen
- The process is called **photolysis**
- Hydrogen is used in dark stage
- Oxygen is released into the atmosphere through the stomates
- Light is also used to make energy in the form of ATP

Dark stage

- It also takes place during the day but it does not require sunlight
- This stage uses carbon dioxide, hydrogen and ATP energy from the light stage
- The carbon enters the leaf through stomates

- Using the ATP energy, the carbon dioxide can be fixed with hydrogen to make glucose
 - The process is called **carbon dioxide fixation**
 - Glucose molecules can be combined to make **starch** in a process called **condensation**
-
- Glucose is the main product of photosynthesis.
 - Glucose cannot be stored by the plant because glucose is very reactive

A TEST FOR STARCH

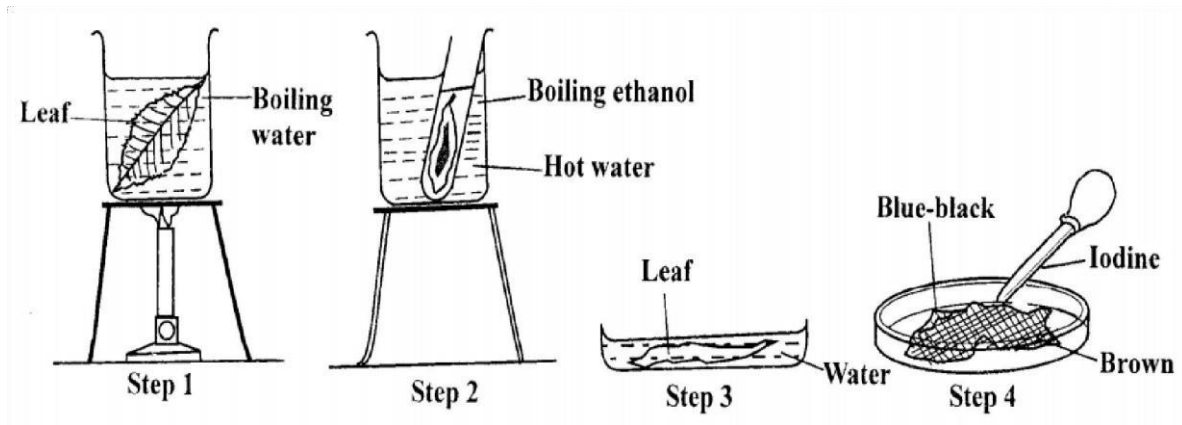
- Pluck a leaf from a green plant
- Destarch the leaf to remove the starch which may be already in the leaf
- Dip the leaf in boiling water for about 30 seconds.
- This stops further chemical reactions by killing leaf cells and destroying enzymes.
- It also makes the leaf more permeable to iodine.
- Boil the leaf in ethanol using a water bath.
- Ethanol dissolves chlorophyll and the leaf looks white. Ethanol is not heated directly because it is flammable.
- Ethanol makes the leaf hard and brittle.
- To soften the leaf, dip it in warm water.
- This also removes ethanol from the leaf.
- Spread the leaf on a white tile.
- This makes any colour changes visible.
- Place a few drops of iodine on the leaf to see colour changes.

Results

- Blue-black colour forms or brown colour (colour of iodine) forms

Conclusion

If blue-black colour forms then starch is present. If brown colour appears then there is no starch.



IMPORTANCE OF PHOTOSYNTHESIS

- Food production:
- Oxygen production:
- Reduction of carbon dioxide from the atmosphere
- Source of energy stored in fossils
- It makes carbon available to plants and animals

Factors Affecting the Rate of Photosynthesis

The rate of photosynthesis is how fast photosynthesis occurs. The factors that affect the rate of photosynthesis are:

Light Intensity

- In the dark, photosynthesis is impossible.
- If light intensity is slowly increased, the rate of photosynthesis increases

Carbon dioxide concentration

- An increase in carbon dioxide concentration increases the rate of photosynthesis.
- A decrease in carbon dioxide concentration results in reduction in rate of photosynthesis.

Temperature

- The rate of photosynthesis increases with increasing temperature up to a point.
- If the temperature is too high photosynthesis stops because the enzymes involved in photosynthesis are denatured (destroyed).

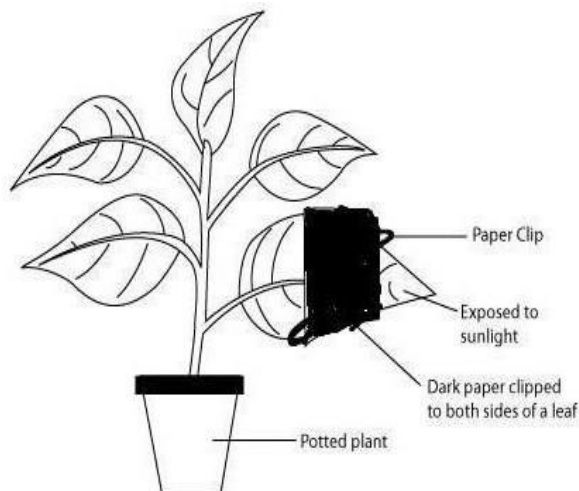
Chlorophyll concentration

- High chlorophyll concentration increases rate of photosynthesis.
- Low chlorophyll concentration decreases rate of photosynthesis.

EXPERIMENTS ON CONDITIONS NECESSARY FOR PHOTOSYNTHESIS

- Before doing experiments on photosynthesis a plant is destarched by placing it in the dark for at least 24 hours.
- In the dark starch is changed to glucose and carried away from the leaf to other parts of the plant, a process called **translocation**.
- Destarching is done to make sure that if glucose is found, should be the one made during the experiment

EXPERIMENT 1: IS LIGHT NECESSARY FOR PHOTOSYNTHESIS?



Materials

- Destarched potted plant
- Iodine solution and ethanol
- Test tube and beaker with water
- Piece of black paper or aluminium foil
- Dropper and white tile
- Bunsen burner **Procedure**

- Take a potted plant and put it in the dark for 24 hours to destarch it.
- Pluck one leaf and test it for starch to make sure the plant has been destarched.

- Cover part of both sides of the leaf using black sellotape or aluminium foil.
- Leave the plant in the sun for about 5 hours.
- Pluck the leaf and remove the aluminium foil and test the leaf for starch.

Results

- Areas which received light turn blue-black with iodine. The part which was covered is brown.

Conclusion

- Light is necessary for photosynthesis.

EXPERIMENT 2: IS CHLOROPHYLL NECESSARY FOR PHOTOSYNTHESIS?

Materials

- Variegated potted plant
- Iodine solution
- Test tube and beaker with water
- Dropper and white tile
- Bunsen burner **Procedure**

- Destarch the plant by placing it in the dark for 24 hours
- Place the plant in the sun for at least 5 hours.
- Take the variegated leaf from the plant and test it for starch.

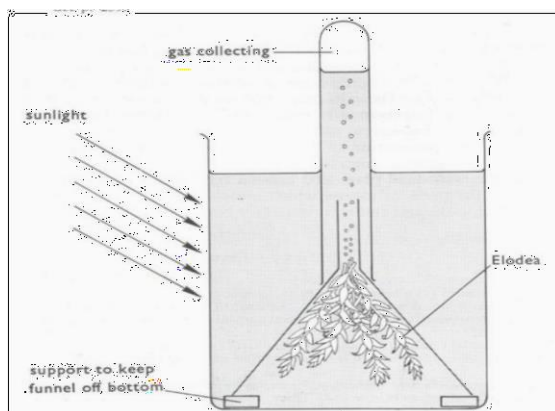
Results

- The part which was green turns blue black in colour and white part is stained brown.

Conclusion

- Since starch is present only in parts which were green, then we conclude that chlorophyll is necessary for photosynthesis.

EXPERIMENT 3: IS OXYGEN PRODUCED DURING PHOTOSYNTHESIS?



Materials

- A water plant (such as Elodea)
- Iodine solution
- Glowing splint
- Test tube and beaker with water
- Dropper and white tile
- Bunsen burner

Procedure

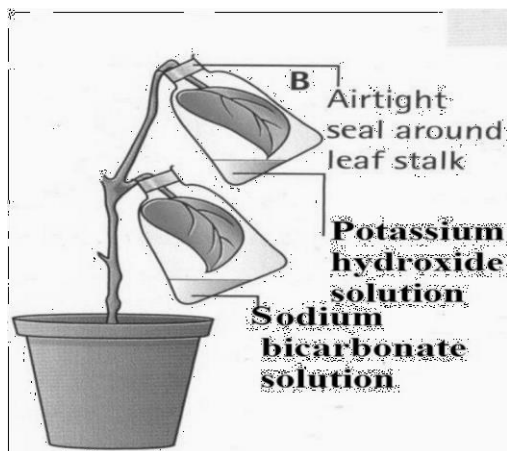
- Place a funnel over the water plant in a beaker containing sodium hydrogen carbonated water
- Invert the test tube over the funnel stem. The funnel should be raised to allow circulation of water
- Place the set up in sunlight
- Bubbles of gas appear rising and collecting in the test tube.
- After sometime the test tube is removed and a glowing splint is inserted.

Result

- The glowing splint bursts into flames.

Conclusion

EXPERIMENT 4: IS CARBON DIOXIDE NEEDED FOR PHOTOSYNTHESIS?



Materials

- Potted plant

Potassium hydroxide

- Sodium bicarbonate
- Iodine solution
- Test tube and beaker with water
- Dropper and white tile
- Bunsen burner

Procedure

- Leave the plant in the dark for 24 hours.
- Choose two leaves on the plant.
- Put one leaf in a flask containing potassium hydroxide solution. Potassium hydroxide solution absorbs or removes carbon dioxide. Leave the leaf attached to the plant.
- Put another leaf in a flask containing sodium bicarbonate solution. Sodium bicarbonate produces extra carbon dioxide.
- Place the plant in sunlight for 5 hours.
- Pluck the chosen leaves and test them for starch.

REVIEW QUESTIONS

1.
 - a. What is photosynthesis?
 - b. Mention any two conditions necessary for photosynthesis
 - c. List down any two products of light stage of photosynthesis
 - d. Give any three functions of the leaf to a plant

2. Give any three reasons for importance of photosynthesis
 - b. Describe how you can test for starch in a leaf
 - c. Describe an investigation you can carry out to show that light is necessary for photosynthesis
 - d. Describe an investigation that you can carry out to show carbon dioxide is necessary for photosynthesis

TOPIC 3: BLOOD DONATION

- Blood donation is a process whereby a healthy person gives blood for use in blood transfusion to a patient.

Blood Transfusion: a process where by a patient is given blood that has been donated by another person

The Need for Blood Transfusion

- Excessive loss of blood by mothers during child birth
- Excessive loss of blood due to an accident or injury
- Acute anaemia in children due to malnutrition and infections
- Excessive loss of blood by patients who undergo surgery

Patients suffering from diseases such as malaria that leads to acute anaemia.

- NB:

The blood collected from donors is stored in hospital stores called **blood banks**.

TYPES OF BLOOD DONORS

a. Family Replacement Donors

- These are family members of a patient who are requested to donate blood.
- Family members donate blood to replace the blood used by their relative (patient).

b. Voluntary Non-remunerated Donors

- These are people who donate blood with desire to save lives.
- Voluntary means the person donates blood willingly without being forced or enticed.
- Non-remunerated means the person donates blood without being paid.

c. Paid or Remunerated Donors.

- These are people who give blood for payment.
- These donors are not recognized in blood transfusion because their aim is not to save lives.

The Process of Blood Donation

The process of blood donation follows the following steps:

- 1) Pre-donation counselling
- 2) Eligibility criteria procedures

3) Venipuncture

4) Post-donation counseling

PRE-DONATION COUNSELLING

- Before donation the donors are taken through a session where they are told about importance of blood donation, Requirement for blood donation, they are also told their blood would be screened and they would be given the results of the tests.
- Other issues discussed include:
 - How to overcome the fear of needles
 - Donors are advised to take a lot of fluids before donating blood
- The donors are also informed on the questions to be asked before blood is donated such as:
 - The history of the donor's family to know whether there are in family with blood diseases
 - Any medication the donor is currently using
 - The health history of the donor
 - The places the donor has visited few weeks to the time of blood donation.

ELIGIBILITY CRITERIA PROCEDURES

At this stage it is determined whether the donor is suitable to donate blood or not.

The criteria used for determining a blood donor include:

The age of the donor

- The donor must be aged between 16 to 65 years. The donor must not be too old or too young because their bodies take a long time to replace the blood donated.

Anaemia

- Individuals with anaemia are not allowed to donate blood.

Sexually Transmitted Infections

- Those infected by sexually transmitted diseases such as HIV & AIDS, Syphilis and gonorrhea are not expected to donate blood.

Prostitutes

- Prostitutes are not recommended to donate blood because they are at high risk of contracting sexually transmitted infections.

VENIPUNCTURE

- Venipuncture is a procedure where a needle is inserted into a vein to obtain blood.
- The donor is sat on a blood donation chair.
- The technician identifies a vein on the arm to carry out venipuncture.
- The technician identifies a vein on the arm. An antiseptic is used to clear the area. A needle is inserted directly into the vein and blood is taken.
- An elastic band may be tied on the upper part of the arm to increase blood pressure in the veins
- a person donates about 450mls of blood in one location every 3-5 months.

POST DONATION COUNSELLING

- After donation the blood is taken to a blood transfusion laboratory where blood is screened.
- Screening means carrying out tests on the donated blood to determine its suitability for donation.
- The blood is tested for HIV, Hepatitis, pathogens (germs) and Rhesus factor
- The results of the tests are confidential.
- If the blood has no problem then the donor is counselled on how to remain healthy.
- if the blood has infections such as HIV, syphilis and hepatitis the person is counselled on how to live positively in case of HIV and to seek medication in case of other infection.

MALAWI BLOOD TRANSFUSION SERVICE (MBTS)

MBTS was established by government with help of European Union in 2003.

Objectives of MBTS

- To reduce incidences of HIV & AIDS and other diseases transmissible by blood
- To provide safe and adequate blood supply to all health care facilities
- To provide adequate supplies of safe blood and blood products to meet the need of all patients in all hospitals in Malawi.

Activities of MBTS

- a. Carrying out district campaigns
- b. collecting blood from donors
- c. Testing and screening the collected blood
- d. Supplying safe and adequate blood and blood products to authorized hospitals.
- e. encouraging blood donations among students, the youth and communities

- f. Encouraging establishment of club 25. Its members commit to donate blood at least 25 times during their lifetime. They also assist the MBTS to encourage others to donate blood.
- g. operating three static clinics in all the three regions of Malawi
- h. operating mobile clinics

Revision Exercise

1.
 - a. Define blood donation
 - b. Explain two reasons for need of blood transfusion
 - c. Describe the types of blood donors
2.
 - a. Describe the stages involved in the blood donation process
 - b. State any two eligibility criteria for blood donors
 - c. Mention any two aims of the Malawi Blood Transfusion Service
 - d. Describe any two activities of the Malawi Blood Transfusion Service

TOPIC 4:THE HUMAN RESPIRATORY SYSTEM

- Respiration is the breakdown of organic food substances to release energy.
- Respiration takes place in cell organelles called **mitochondria**.
- Mitochondria are found in almost all cells but they are abundant in cells that need a lot of energy such as sperms and muscle cells.

Breathing: the taking in of oxygen and release of carbon dioxide by animals.

Gaseous exchange: the exchange of oxygen and carbon dioxide across a membrane.

Importance of Breathing

Allows organisms to get oxygen for respiration in the cells

TYPES OF REPIRATION

Aerobic respiration

Anaerobic respiration

AEROBIC RESPIRATION

- This type of respiration takes place in the presence of oxygen.
- Glucose reacts with oxygen and produces carbon dioxide, water, and energy.
- Energy is the **main product**; **water** and carbon dioxide are **by-products**.

Glucose + Oxygen $\longrightarrow$ Carbon dioxide + Water
+ Energy



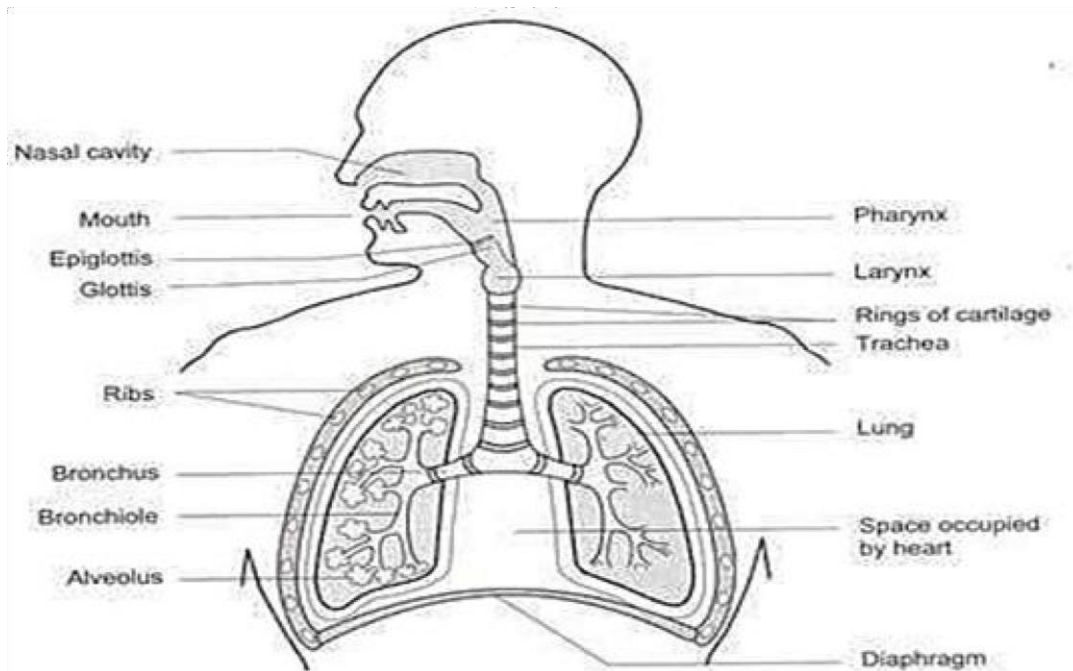
ANAEROBIC RESPIRATION

The release of energy from organic food substances in the absence of oxygen. In plants and fungi it is called **alcoholic fermentation**. Because alcohol or ethanol is produced.

Glucose $\longrightarrow$ ethanol + Carbon dioxide + Energy

$C_6H_{12}O_6 \longrightarrow 2CH_3CH_2OH + 2CO_2 + 210kJ$

Energy is the main product, and **Carbon dioxide** and **ethanol** are **by-products** of anaerobic respiration in plants.



- In animals like human muscles anaerobic respiration is called **lactic fermentation** because acid is produced

In animals, anaerobic respiration produces **energy** as a **main product** and **lactic acid** as a **by-product**.

Properties of efficient Gaseous Exchange Structures

- a. They are permeable to oxygen and carbon dioxide
- b. They are moist-to dissolve the gases so that they are transported by blood in animals with transport systems
- c. They are thin for faster diffusion due to short distance
- d. They are rich in blood capillaries in animals with transport systems to transport gases.
- e. They have large surface area for maximum gaseous exchange

GASEOUS EXCHANGE IN HUMANS

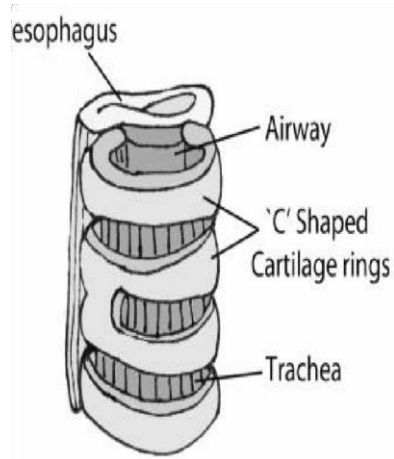
Gaseous exchange in humans takes place in the alveolus located in the lungs.

Structure of the Human Respiratory System

- Twelve pairs of ribs surround and protect the lungs and the heart.
- Intercoastal muscles are attached to the ribs.
- A large sheet of muscle (diaphragm) separates the thorax from the abdomen

The nostrils/nasal cavity: have hairs that filter the air. It also warms and moistens the air.

The pharynx: it is located behind the nasal cavity and above the larynx. It is a food and air



passage.

Trachea: made of rings of cartilage that keeps the trachea open.

Rings of cartilage prevent the collapse of the trachea during inhalation.

Larynx/Voice box: it is a passage of air. It has two membranes called vocal cords. The vocal cords can be made to vibrate to produce speech.

Bronchi: these are branches of the trachea. They branch into bronchial tubes which further divide into bronchioles that lead to the alveoli.

Alveoli: they are a site of gaseous exchange in lungs.

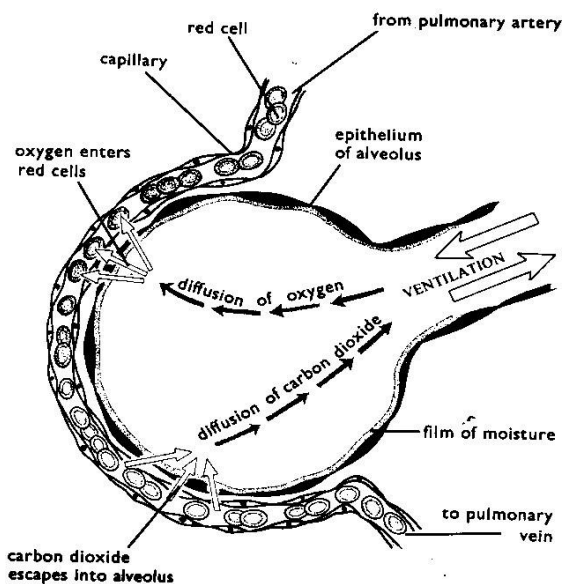
Diaphragm: a dome shaped sheet of muscles located at the bottom of the rib cage.

During inhalation it contracts and moves downwards and increases volume of chest cavity and air rushes into lungs.

During exhalation it relaxes and air is expelled.

Intercoastal muscles: during inhalation the external intercoastal muscles contract and pull the chest cavity upwards and outwards. This increases volume of the chest cavity and air rushes into the lungs. During exhalation internal intercoastal muscles contract pulling the chest cavity inwards and downwards and air rushes out.

The alveolus



This is the site of gaseous exchange in lungs.

- Oxygen diffuses through a thin epithelium of the alveolus and the endothelium of the capillaries.
- It combines with haemoglobin to form **oxyhaemoglobin**.
- Carbon dioxide diffuses in reverse direction.

ADAPTATION OF THE ALVEOLUS FOR GASEOUS EXCHANGE.

- a. It has a network of blood capillaries that carry the gases being exchanged.
 - b. Thin walls in between capillaries and alveolus – gases diffuse faster.
 - c. It is lined with a thin film of moisture to dissolve the diffusing gases.
 - d. Diffusion gradient: There are differences in concentrations of gases in the blood and alveolus, more oxygen in alveolus than the blood hence oxygen diffuses into capillaries, and more carbon dioxide in blood capillaries hence oxygen diffuses into the alveolus.
- They are numerous: increase surface area for gaseous exchange.
 - The lung is surrounded by a **pleural cavity**. It is a space lined by pleural membranes, membranes protect the lungs, stop them leaking air into the thoracic cavity.

- In between the membranes there is **pleural fluid** that reduces friction between lungs and thorax.

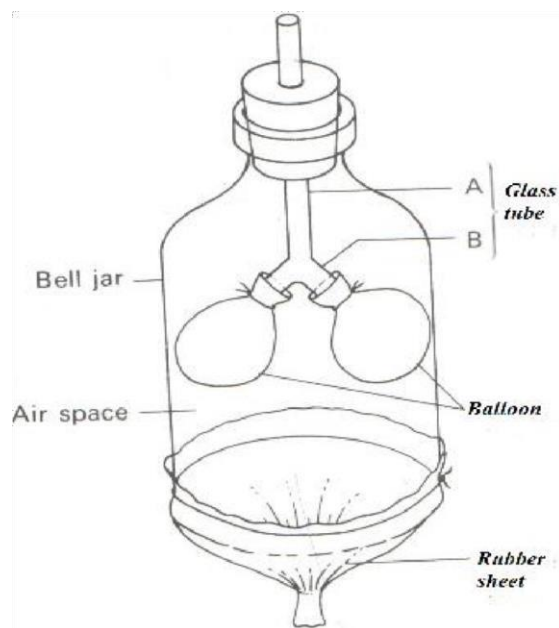
Activity: To Demonstrate the Mechanism of Breathing Using Lung Model

Materials

- Bell jar
- Two balloons
- Rubber stopper with a hole
- Glass tube
- Rubber band

Procedure

- Assemble the materials as shown in the figure below



Study the model and state what the following parts represent

In the respiratory system – bell jar (rib cage)

- balloon (lungs)
- rubber sheet (diaphragm)

- air space(chest cavity)
- glass tube (trachea/bronchi)

Pull the rubber sheet at the bottom of the bell jar. What happens?

- Now push up the rubber sheet. What do you observe? **Discussion**
- Pulling at the rubber sheet causes the balloons to inflate while pushing the sheet causes the balloons to deflate. This represents exhalation and inhalation.

Limitations of the lung model

- The bell jar which represents the rib cage does not move to change the volume of the air space inside the glass jar.

Breathing mechanism

Breathing is the taking of air into and outside the lungs.

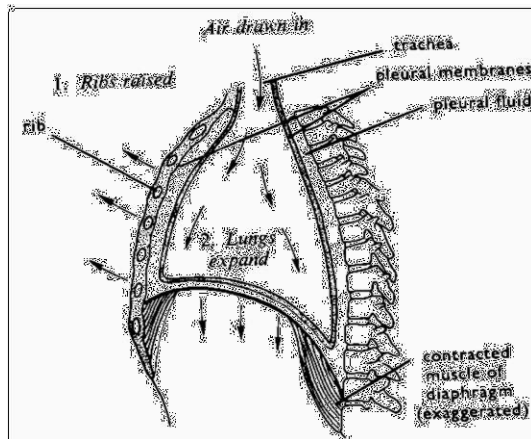
The following structures help in breathing:

- Ribs
- Diaphragm
- Intercostals muscle (both external and internal)

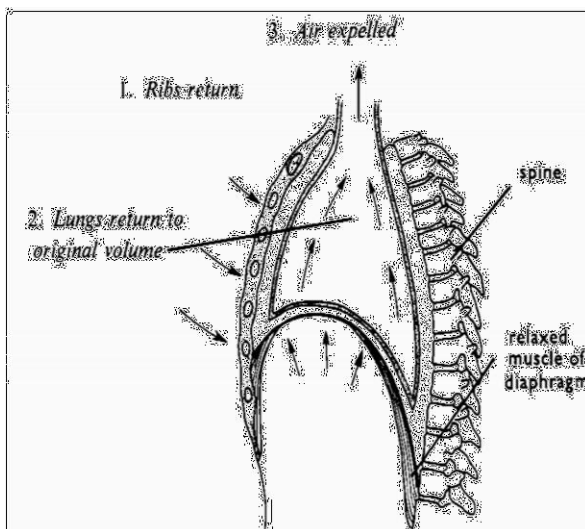
Breathing takes place in two phases:

- Breathing in (inspiration)
- Breathing out (expiration)

BREATHING IN (INHALATION)



- Diaphragm contracts and moves downwards (flattens).
- External intercostal muscles contract and pull the rib cage upwards and outwards.
- This increases the volume of the chest cavity. Air pressure inside the chest cavity and that of the lungs decreases compared to atmospheric air pressure.
- Rushes into the lungs from the atmosphere.



BREATHING OUT (EXHALATION)

- Diaphragm relaxes and returns to its dome shape.
- External intercostal muscles relax but internal contract and pull rib cage downwards and inwards.
- Lung volume decreases and pressure increases. Air rushes out.

Regulation/control of Breathing.

- Breathing is controlled by the Medulla oblongata.
- Medulla oblongata monitors the levels of carbon dioxide and oxygen in the blood stream and adjusts the breathing rate to maintain balance of the gases.
- In case of high carbon dioxide concentration receptors in the Aorta and carotid arteries detect this.
- Receptors send an impulse to the oblongata which sends messages to the breathing structures such as lungs to increase air intake and transport organs such as the heart to speed transport of gases.

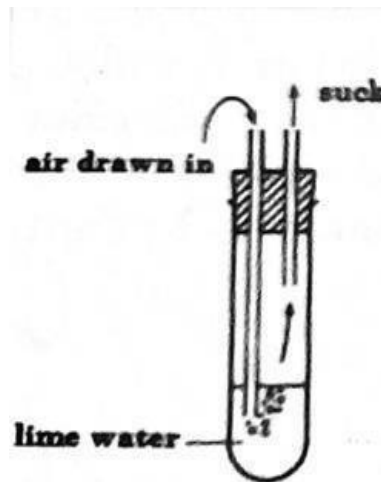
Comparison of Inhaled and Exhaled Air

	% in inhaled air	% in exhaled air
Oxygen	21	16
Carbon dioxide	0.03	4
Nitrogen	79	79
Water vapour	Varied usually less	Saturated
Heat/Temperature	Usually Less	Body temperature

Testing for Carbon Dioxide inhaled air and Exhaled air

Inhaled Air

- Set up the test tube as shown below.



- Suck the through the short tube.
- Make sure you are using the tube that has not reached the lime water.
- Suck the air for some time and observe what happens.

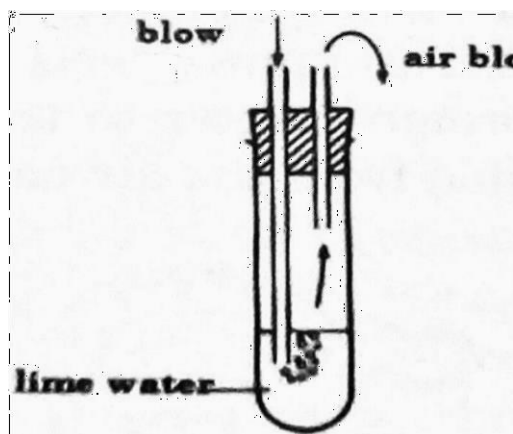
Results

After a long time the lime water turns milky white

This is because inhaled air contains less carbon dioxide than exhaled air.

Exhaled Air

- Set up the test tube as shown below



- Blow the air through the longer tube that touches the lime water for sometime
- Observe what happens **Result**
- The lime water turns milky white faster than in inhaled air because it contains more carbon dioxide than inhaled.

Investigating the Presence of water in Exhaled water

- Take a mirror and place it before your face
- Take a deep breath and then blow out air over the mirror several.
- Wait for few minutes.
- You will see the mirror becoming cloudy.
- This shows that water vapour has condensed on the mirror.
- Take cobalt chloride paper onto the mirror. Observe what happens.

Result

- The cobalt chloride turns pink in presence of water
- This shows that exhaled air contains water.

Temperature of Exhaled and Inhaled Air

a. Using a thermometer

- Use the ordinary thermometer to take temperature of the surroundings.
- This is the temperature of the inhaled air.
- Record the readings

- Clean the thermometers with a disinfected cloth.
- Put the bulb of the clinical thermometer in the space between the tongue and the roof of the mouth to measure temperature of exhaled air.
- Wait for two minutes. Take the temperature readings and record
- Compare the readings of ordinary thermometer to those of the clinical thermometer

Result

- The ordinary thermometer will record lower temperature than clinical thermometer.
- This shows that exhaled air has higher temperature than the inhaled air.

b. The feel Method

- Wave your hand and feel the air around
- Blow the air from your mouth onto your hand.
- Make sure your hand is close enough to your mouth.
- Compare the difference in feeling

Result

- The exhaled air feels warmer than the surrounding air.
- This means exhaled air has higher temperature than inhaled air

ARTIFICIAL RESPIRATION OR VENTILATION

- This is the first aid practice carried out on a person who is having difficulties in breathing due to an accident, fainting or health complications.
- It is also called **artificial resuscitation**.
- Artificial ventilation is done by blowing air using one's mouth into the mouth of the patient.
- The person carrying out the first aid places their mouth onto the mouth of the patient and blows air into the patient.

- This makes artificial resuscitation to be called **kiss of life**.

Procedure for carrying out Artificial Resuscitation

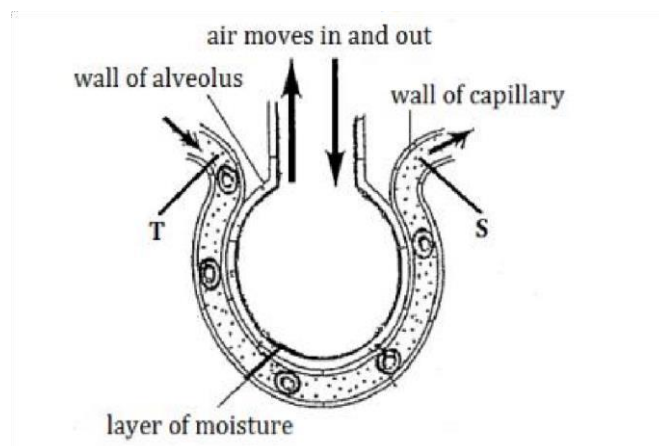
- Let the patient lie on his or her back on a mat or soft material
- Close the patient's nose and pull the head backwards
- Take a deep breath then place your mouth over the patient covering it completely.
- Breathe out heavily forcing the air into the lungs of the patient
- Remove your mouth and gently press the chest of the patient to force air out.
- Repeat the fourth and fifth procedure until the patient starts to breathe without help
- If the breathing fails after 20 trials, rush the patient to the nearest facility

The Debate on the Mouth to Mouth Method

- In the past mouth to mouth resuscitation used to involve direct contact between the patient and the first aider.
- Nowadays, a handkerchief is used to cover the mouth during resuscitation to avoid infections
- There is still debate on whether mouth to mouth is still safe.

Revision Exercise

1. a. State any two importance of breathing
b. Define respiration
c. Name the part of the lungs which is the site for exchange of gases
2. a. State any three properties of a gaseous exchange structure



- b. Describe any three ways in which the alveolus is adapted for exchange of gases
 - c. Give anyone importance breathing through the nose
 - d. State the function of the pleural fluid
 - e. Describe the breathing mechanism
3. a. Describe any two ways in which inhaled air differs from exhaled air b. What is choking?
c. What is suffocation?
4. The diagram below shows an alveolus surrounded by blood capillaries
- a. To which chamber of the heart does blood flow from point S?
 - b. State the difference in concentration of oxygen between blood passing at T and blood passing at S.
 - c. What is the role of the thin film of moisture found in the lining of the alveolus?
 - d. Explain the disadvantage of using mouth to mouth method of resuscitation?

TOPIC 4: HUMAN NERVOUS SYSTEM

- The linking together of various processes in an animal is called coordination.
- In humans and other mammals nervous and endocrine systems bring about coordination.

Definition of Terms

Stimulus: the change in the environment of an organism that influence the behaviour of an organism.

Response: It is a reaction to a stimulus by an organism

Irritability: the ability of an organism to respond to a stimulus.

Receptors: these are parts of the body that detect stimuli such as eyes, ears, nose and skin and tongue.

Effectors: these are parts of the body that respond to a stimulus.

Impulse: electrical message or information transmitted by nerve cells or neurons.

Neuron: a specialized cell that conducts impulses.

Types of Stimuli

- There are two types of stimuli: external and internal stimuli

External Stimuli

- These are the changes in the external environment of an organism which are perceived by an organism.
- Examples of external stimuli include sound, light, temperature, touch and smell.

Internal stimuli

- These are within the body of an organism.
- Examples include body temperature, salt and sugar concentration, blood pH and water levels.

Types of Animal Responses to Stimuli

- Chemotaxis:** This is animal responses to chemicals. Animals can move towards or away from chemicals.
- Photo taxis:** This is animal responses to light. Animals can move towards or away from light.
- Hydro taxis:** This is animal responses to water or moisture.
 - Animals can move towards or away from water or moisture.
- Chemotaxis:** This is animal responses to heat. Animals run away from strong heat.
- Haptotaxis:** This is animal responses to touch.

Importance of Animal Responses

- They enable animals to locate food such as responses to chemical substances.
- Enable organisms to move away from danger.
- They enable organisms to move to environments with favorable conditions for their survival.
- They enable organisms to locate mates during the breeding seasons.

THE NERVOUS SYSTEM

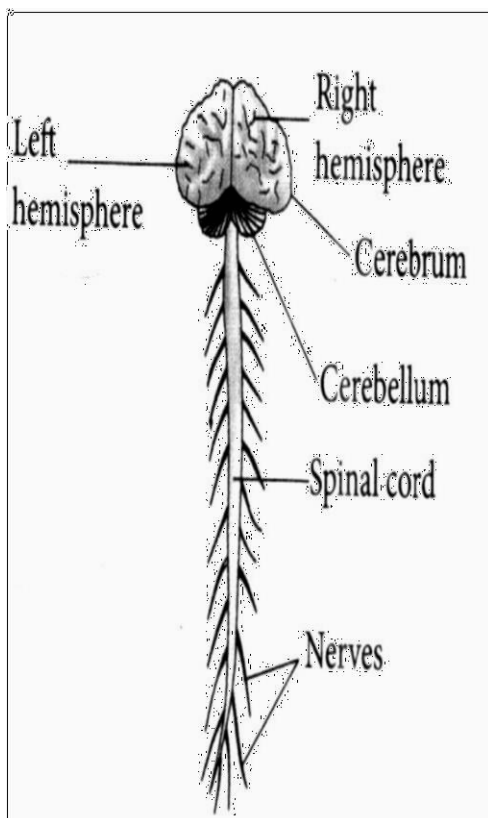
- The nervous system of any mammal consists of:

- i. **The central nervous system (CNS):** this consists of the brain and the spinal cord
- ii. **The peripheral nervous system:** this consists of a branching system of nerves linking the brain (cranial nerves) and spinal cord (spinal nerves) with all parts of the body.
- iii. **The autonomic nervous system** consisting of centres that control life sustaining reflexes such as breathing, heartbeat, etc.

Functions of the nervous system

- Receiving stimuli from the environment.
- Converting the stimuli into electrical impulses, a process called transduction.
- Transmitting impulses over distances.
- Interpretation of impulses and coordination of responses to stimuli.

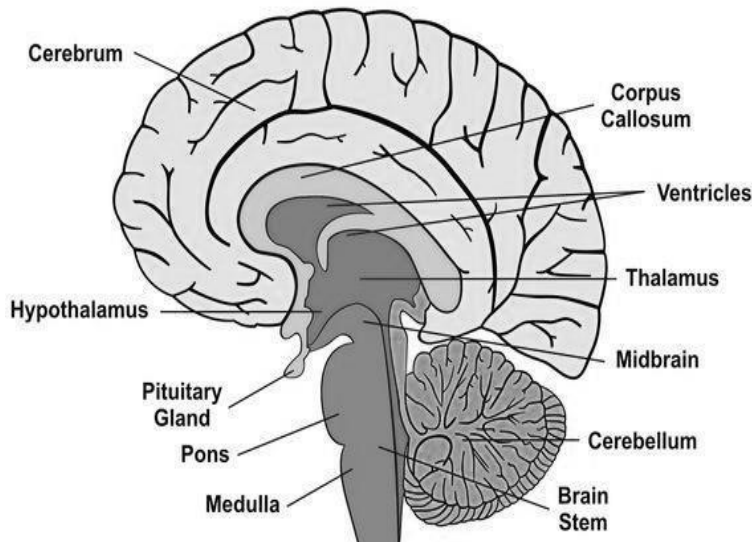
THE CENTRAL NERVOUS SYSTEM



- The brain is made up of two halves called hemispheres (the right and left hemispheres).
- The hemispheres are interconnected by a group of nerves called **corpus callosum**.
- The right hemisphere controls activities of the left side of the body while the left hemisphere controls activities of the right side of the body.

- The outermost part of the brain is grey due to cell bodies and innermost part is white due to axons. The brain is covered by two membranes called meninges.

Functions of the Major Parts of the Brain



1. Cerebrum

- This is the largest part of the human brain. It is divided into right and left cerebral hemispheres.
- Its outer layer is called cerebral cortex or grey matter.
- It is highly folded hence higher number of neurones.
- The large surface area enables humans to carry out complex activities.

Functions of the Cerebrum

- It is the memory centre
- It is involved in learning. iii. It is the intelligence centre.
- It is involved in creativity and imagination.
- Contains centres for all five senses which receive and interpret impulses from sense receptors.

2. Cerebellum

- The cerebellum is located below the rear part of the cerebrum.
- It is smaller than the cerebrum. It is also highly folded. **Functions of the Cerebellum**

- Coordination of body movements (muscular coordination).
- Maintains balance and posture.

Ensures dexterity in fine movements like playing the guitar, sewing and typing.

3. Medulla Oblongata

This is the hind most and smallest part of the brain.

Functions of the Medulla Oblongata

- It controls vital reflexes that regulate heartbeat rate, breathing, blood circulation and blood pressure.
 - It controls non-vital reflexes that coordinate swallowing, salivation, coughing and sneezing.
 - The following are the general functions of the brain
- It receives and interprets impulses from all sensory organs of the body.
 - It sends off impulses to glands and muscles causing them to function as effectors.
 - It stores information so that behaviour can be modified according to past experience.

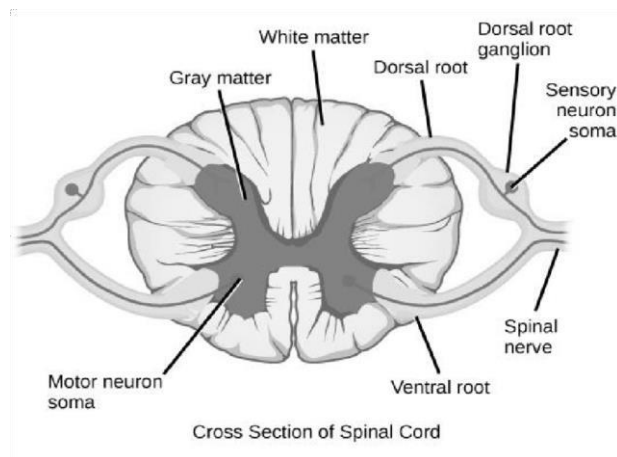
4. The Hypothalamus

- It performs the following functions:
- It controls secretion of hormones by the pituitary glands.
 - It regulates some homeostatic processes such as carbon dioxide concentration and water levels in the blood.
 - It controls hunger
 - It controls sleep
 - It also controls thirst

THE SPINAL CORD

- The spinal cord is an extension of the brain running from the base of the brain down the back.
- It is protected by the vertebrae of the backbone and meninges.
- The outer parts of the spinal cord contain the white matter and the inner part contains the grey matter.

Cross-section of the Spinal Cord



There are 31 pairs of spinal nerves and these divide form two branches, the dorsal and ventral roots.

Dorsal root: also known as the sensory root is made of sensory neurons that bring impulses from receptors.

Ventral root: also known as motor root is made of motor neurons that carry impulses from the spinal cord to effectors.

Ganglion: is a bulge made by cell bodies sensory neurons in the dorsal root.

- All cell bodies apart from those of the dorsal root are concentrated in the grey matter.
- The white matter consists of nerve fibres.

White matter: it covers the peripheral areas of the grey matter. The white matter consists of nerve fibres that carry impulses up and down the spinal cord.

Grey matter: this is a butterfly shaped that appear grey in fresh tissue. It is grey due to presence of numerous cell bodies.

Central/spinal canal: it is a small clear space. It is filled with **cerebrospinal fluid**.

Functions of the Spinal Cord

- i. It links nerves of the peripheral nervous system with the brain.
- ii. Coordinates certain automatic or reflex responses.

DRUGS AND THE CENTRAL NERVOUS SYSTEM

- A drug is a chemical substance that can alter the way in which a part of the body works.
- Drug abuse is the use of drugs in a way that can cause harm to the person using them or someone else. Drugs affecting the brain and spinal cord fall under the following groups:

Sedatives/depressants: these drugs act by reducing brain cell activity hence reduce anxiety and induce sleep. Examples of sedatives include valium, alcohol and opium.

Stimulants: these speed up the action of the nervous system.

Stimulants make the drug user more confident and alert. Examples include caffeine and nicotine.

Pain killers (analgesics): they suppress the part of the brain or spinal cord responsible for pain or inhibit formation of impulse at the site of pain.

Examples of pain killers include aspirin, morphine and ibuprofen.

Hallucinogens: these drugs cause hallucinations or illusions.

Hallucinations are perceptions of somebody or something that is not there.

Examples of hallucinations include

Drugs like marijuana can lead to total breakdown of mental coordination leading to madness.

Effects Drugs on the Nervous System(Alcohol and Indian hemp)

- a. Alcohol and marijuana can slow down the transmission of impulses which slows down the speed at which a person can respond to a stimulus.
- b. Can damage brain cells. This leads to loss of memory.
- c. Drugs can also damage cells that control emotions.
- d. This results to distorted emotional reactions by an individual.
- e. This leads to slow or no emotional reactions in situations where it is needed and feelings of worry and anxiety even when there is nothing to worry about.

Revision Exercise

1. Define the following terms
 - a. Stimulus
 - b. Receptor
 - c. Chemotaxis
 - d. Hydro taxis
2. Mention the three main parts of the nervous system of a mammal
3. Mention the two main parts of the central nervous system
4. Describe the following groups of drugs
 - Sedatives
 - Hallucinogens
 - Stimulants

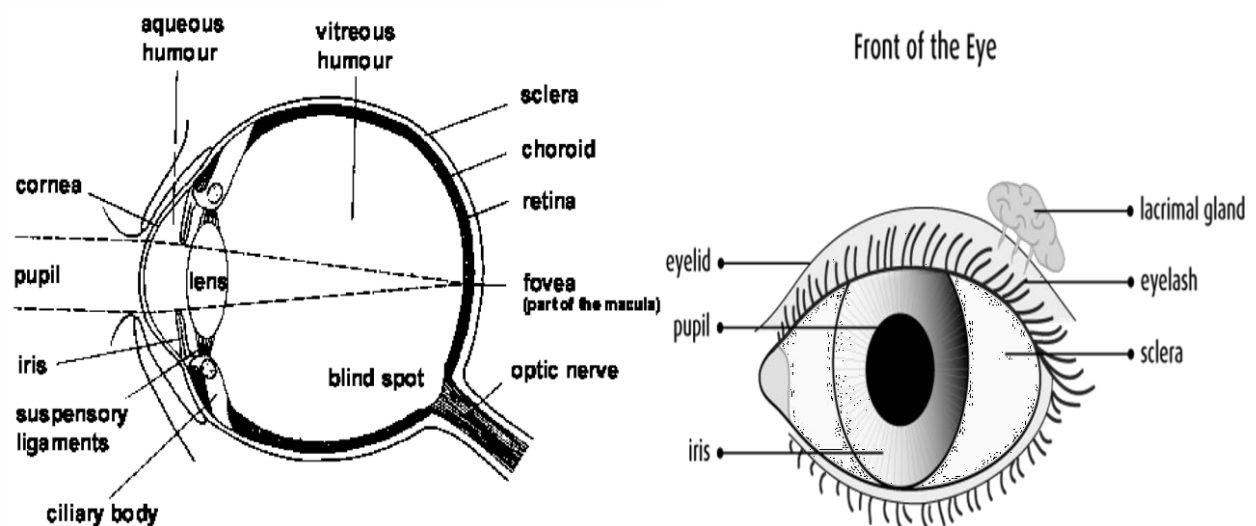
TOPIC 5: SENSE ORGANS

- Sense organs are parts of the body that detect a stimulus. Sense organs are also called **receptor organs**.

Sense Organ	Stimulus
Eye	Light
Ear	Sound
Tongue	Taste
Nose	Smell
Skin	Temperature, pressure and touch

- The sense organs play a very important role in body of an organism as shown in the table

THE EYE



- It is the sense organ that enables us to see

It consists of the eyeball located in a spherical space or hollow in front of the skull called the orbit or eye socket.

Parts of the Eye and Their Functions

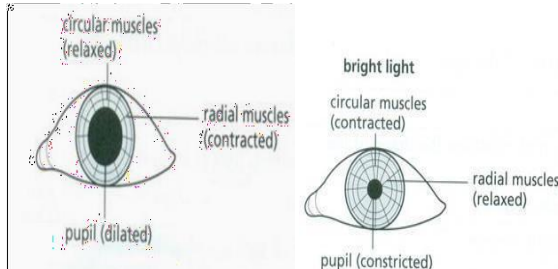
On the outside of the eye there are eyelids that cover and protect the eye.

The edges of the eyelids secrete tears.

Tears perform the following functions:

- a) They keep the eye moist
 - b) They wash away dust particles
 - c) They contain an enzyme called lysozyme that kills and digests germs
- .
- Excess tears drain away into the nasal cavity through a duct called the tear duct.
 - The eyelashes prevent small particles from entering the eye.

Part of the Eye	Description	Functions
<i>Sclera (sclerotic coat)</i>	The white outer layer of the eyeball. It is inelastic and Opaque.	It protects the inner delicate Parts of the eyeball.
<i>Cornea</i>	The transparent front part of the sclera.	Allows light to enter the eye. It also refracts (bends) the light.
<i>Conjunctiva</i>	Thin transparent skin or membrane covering the cornea.	It also protects the eyeball
<i>Choroid</i>	The deeply pigmented layer which lines the inside of the sclerotic coat. It contains cells that have melanin, arteries and veins.	It prevents reflection of light within the eyeball. It also supplies oxygen and nutrients to the eye due to presence of the blood vessels

<i>Ciliary body</i>	It is a ring of thickened tissue which is continuous with the choroid. It also contains the ciliary muscle.	It produces the aqueous humour that maintains the shape of the eyeball. The ciliary muscles help in accommodation
<i>Iris</i>	Part of the choroid coat at the front of the eye. Contains melanin, circular and radial muscles.	It regulates the size of the Pupil. Melanin gives the eye its colour.
<i>Pupil</i>	Hole at the centre of the iris 	It regulates the amount of light entering the eye. It gets wider in dim light and narrows in bright light due to contraction and relaxation of ciliary muscles.
<i>Lens</i>	Transparent biconvex disc	Focuses (refracts) light rays to form an image on the retina. Involved In accommodation by change in thickness.
<i>Suspensory ligaments</i>	Fibrous ligament attached to the ciliary body	Attaches the lens to the ciliary muscle or holds the lens in position

<i>Aqueous humour</i>	Transparent watery fluid in front of the lens	It contains nutrients for the cells of the eye tissue since the cornea and lens have no blood supplies. It also maintains shape of the eyeball. It also Refracts light.
<i>Vitreous humour</i>	Transparent jelly behind the lens	It fills the eyeball and maintains its shape. It supplies nutrients to eye tissue. It also Refracts light.
<i>Retina</i>	Transparent, contains light sensitive cells (rods and cones)	Rods sense dim light while Cons sense colour and brightlight.

Study the problems of the eye and how they can be treated.

Blindness

- Blindness occurs in various forms which include colour blindness, night blindness and total blindness

Colour blindness this is inability to distinguish between red, green and yellow colours.

- It is inherited from parents.
- It is more common in males.

Night blindness

- This is the inability to see properly in dim light.
- It is usually caused by lack of vitamin

Total blindness this is when there is a total loss of sight.

- This may be caused by diseases such as trachoma, syphilis or gonorrhea or by physical damage to the eye and its nerves.

d. Astigmatism

- It is caused by uneven or irregular surface of the lens or cornea.
- Sometimes the cornea may have rough and uneven surfaces.
- As a result of this, light passing through them are bent at different angles and are scattered and not all of them are focused on the retina.
- This distorts images falling on the retina.
- Part of the image may be out of focus does not pass through.
- This is caused by ageing, exposure to intense heat or atomic radiation.
- Cataracts can be corrected by surgically the opaque lens and replacing it with a clear plastic lens.

f. Glaucoma

- Glaucoma is most common in older people.
- It is caused by increased pressure inside the eye due to buildup of aqueous humour.
- This damages the optic nerves causing blindness and great pain.
- The condition can be corrected by surgically making an incision in the sclera to allow the fluid to filter out.

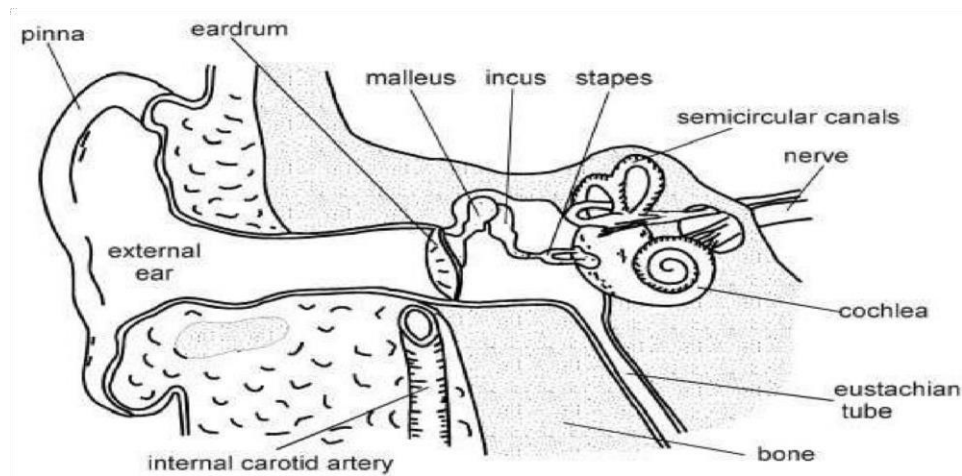
Ways of Caring for Eyes

Avoid reading in very bright light, looking directly at the sun or at very bright light.

Avoid reading under poor light as this may strain your eyes

- If a small particle or insect lands in your eye, do not rub the eye but blink continuously or ~~bathe~~ bathe the eye in clean water.
- See a doctor if the particle does not come out.
- Always wash your eyes with clean warm water.
- When using a computer take a 40 minute break to take the eyes from the computer screen.
- At the same time blink to ensure that the cornea is kept moist.
- Have regular eye check up by an optician

THE EAR



THE OUTER EAR

- The ear has three main parts: the outer ear, the middle ear and the inner ear.
- The outer ear is made of the earlobe or the pinna, auditory canal and the ear drum.

The Pinna

- This is the ear flap on the side of the head.
- It is made of skin and cartilage.
- Collects sound waves and directs them into the auditory canal or ear tube.

The Auditory Canal/tube

- The tube leads sound waves towards the eardrum.
- It is lined with small hairs that help to keep out dust and small insects.
- It is lined with cells that secrete wax.
- This wax keeps the ear drum soft and traps dust and germs.

The Eardrum

- This is a thin membrane that is stretched over the auditory canal.
- It separates the outer ear from the middle ear.

- The eardrum passes sound vibrations from the outer ear to the ossicles in the middle ear.

THE MIDDLE EAR

- The middle ear is an air-filled space chamber.
- It is separated from the outer ear by the ear drum.
- It is also separated from the inner ear by the oval and round windows.
- The middle ear consists of the Eustachian tube, hammer, anvil, and stirrup, which include the oval window and round window.

The Ossicles

- The ossicles consist of three small bones called ~~hammer~~ (malleus), anvil (incus) and stirrup (stapes).
- They are suspended by muscles into the middle ear.
- The functions of the ossicles are:
 - To receive sound vibrations from the eardrum and pass them to the oval window.
 - To amplify the sound vibrations before passing them to the oval window.

The Eustachian Tube

- This is a tube from the floor of the middle ear that opens into the back of the mouth or pharynx.
- It ensures equal pressure on both sides of the eardrum.

The oval Window

- It is the last part of the middle ear.
- It passes the sound vibrations to the inner ear.

THE INNER EAR

- The inner ear consists of cochlea, branches of the auditory nerve and semi-circular canals.

Cochlea

- It is a coiled tube in the inner ear.
- It is responsible for hearing.

Branches of the auditory nerve

- They are to the cochlea.
- They transmit impulses from the ear to the brain.

Semi-circular canals

- These are three canals in the inner ear placed at right angles to each other.
- They contain a fluid.
- The semi-circular canals are responsible for balance.

Functions of the Ear

- It is an organ for hearing
- It helps in maintaining or keeping balance

The Hearing Process

- Sound waves are collected and then travel along the auditory canal.
- The waves cause the eardrum to vibrate.
- The vibrations of the eardrum are transmitted through the ossicles.
- The vibrations are transmitted from the ossicles to the membranes of the oval window.
- The vibrations of oval window membranes cause the fluid in the cochlea to vibrate.
- The sensory cells connected to the cochlea are stimulated by the vibrations to produce impulses.
- These impulses are conducted by the auditory nerve to the brain (cerebrum).
- The brain interprets these vibrations as sound.

Balance

- When the body changes position the fluid in one or more of the semi-circular canals moves.
- The ~~me~~ cells stimulate sensory cells in the canals to produce impulses that are sent to the brain.
- The brain realizes the position of the body and sends an impulse to the muscles concerned to keep the body in that position or to change position.

DEFECTS OF THE EAR

- Deafness is the main defect of the ear.
- Deafness is a condition whereby an individual is unable to hear.

Types of Deafness

a. Absolute or Nerve Deafness

This is a condition whereby sound vibrations are not able to reach the brain.

- Nerve deafness may be due to damaged auditory nerve
- Damaged cochlea making them unable to detect stimulus (sound vibrations)
- Damaged brain cells that are involved in sound reception
- Heredity-a child may be born with some parts of the inner ear missing or not functioning
- In most cases nerve deafness cannot be corrected.
- Affected individuals can be assisted to live normal lives by:
 - Use of visual signs during speech -Learning skills of lip reading
- **Conductive Deafness** this is a condition where by a person is partially unable to hear due to of transmission of sound vibrations to the inner ear.

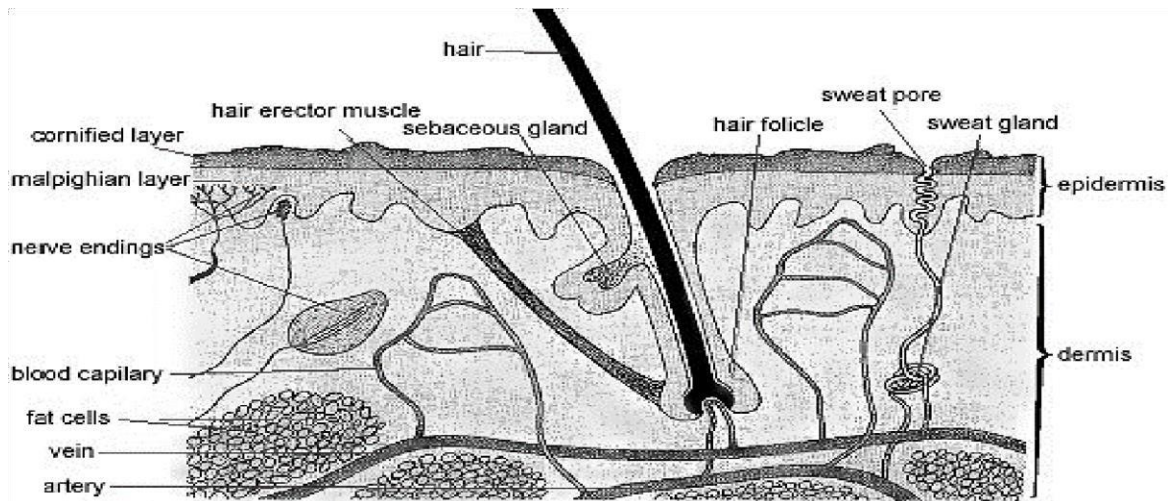
- Conductive deafness may be due to:
 - Accumulation of wax in the ear canal
 - Damage or rupturing of the eardrum by objects, blows on the head and loud sounds
 - Ear infections leading to production of pus that reduces movement in the ear.
 - Too much ossicles growth.
 - This reduces their ability to vibrate hence interfering with transmission of sound vibrations.
- Use of certain drugs in some individuals such as chloroquine in some individuals.

Correction for Conductive Deafness

- Treatment of ear infection by an ear specialist (audiologist)
- Too much bone growth of the ossicles can be corrected by an operation
- Hearing aids can be used to correct conductive deafness
- The wax can be removed by a doctor

Caring for the Ear

- Do not poke the ear with sharp objects such as sticks to avoid damaging the eardrum
- Clean the ear canal with soft cloth with warm water and soap to prevent accumulation of wax
- Protect your ears from loud sounds or noises
- Avoid use of earphones with amplified sound to avoid damaging the eardrums
- Infections should be treated early enough to prevent damage to the eardrum or cochlea
- If an insect enters your ear, do not shine into the ear to look for it.
- This makes the insect to go deeper into the ear.



THE SKIN

- The skin is the largest organ in the body.
- It forms a continuous covering over the surface of the body.
- It allows us to experience touch, pressure, cold and heat.
- The skin consists of two layers: the epidermis and dermis.

THE EPIDERMIS

- The epidermis is the tough, protective outer layer. It is the upper part of the skin.
- It is made of dead cells and is called the **horny layer**.
- It is made of three layers: the cornified layer, the granular layer and the Malpighian layer.

b. The Cornified Layer

- It is the outermost part of the epidermis.
- It is made of dead cells which form a tough outer layer that acts as a barrier against entry of germs.
- It also protects inner cells against mechanical or physical damage.
- It also reduces water loss.

The cells in this layer produce a water proof protein called keratin that strengthens them.

c. **The Granular Layer**

- The granular layer is made of living cells which eventually form the cornified layer.
- It is the middle layer of cells in the epidermis.
- The Malpighian Layer
- This is the innermost layer of cells in the epidermis.
- It is made up of actively dividing cells which are responsible for renewal of the epidermis.
- The cells in this layer contain melanin pigment which contribute to the colour of the skin.
- The more the melanin the darker the skin.
- Melanin protects the skin from dangerous radiation from the sun.

THE DERMIS

- It is located below the epidermis.
- It is thicker than the epidermis.
- It contains several parts such as blood vessels, sweat glands and sensory ~~repts~~

- **Sweat Glands**

- It stores food in form of fat
- It produces vitamin D
- These are coiled tubes which open on the surface of the skin through sweat pores.
- They secrete and release sweat through pores on the skin.
- Sweat consists of water, mineral salts and some traces of urea.
- The liquid that forms sweat is absorbed by sweat glands from blood capillaries supplied to each gland.

- **Blood Capillaries**

These are blood vessels that supply the skin with nutrients and oxygen and remove metabolic wastes.

- **Hair Follicles**

- These are tiny pits in the dermis.
- Hair grows inside the follicle due to addition of cells to it at the bottom of the pit.
- Hair is made of a protein called keratin.

- a. **Sebaceous Glands**

- These are small glands that open into the hair follicle.
- They secrete an oily substance called sebum.
- The sebum makes the hair repel water; it softens the epidermis and kills certain bacteria.

Hair erector muscle

- It is attached between the bottom part of the hair follicle and the epidermis.
- When it contracts the hair follicle stands up right and very small pimples or swellings appear on the skin.
- When it relaxes the hair lies flat on the skin.

Sensory Receptors

These are stimulus related to temperature, touch, pressure and pain.

Fatty layer

It acts as an insulator that prevents excessive heat loss from the body.

Function of the Skin

- It acts as a sense organ for touch, temperature, pain and pressure.
- It protects the body against infection, injury and strong sunshine.
- It prevents water loss from the body
- It gets rid of waste products such as urea, excess water and mineral salts
- It regulates temperature through sweating and changing depth of blood flow in the skin.
- It stores fats which are source of energy.

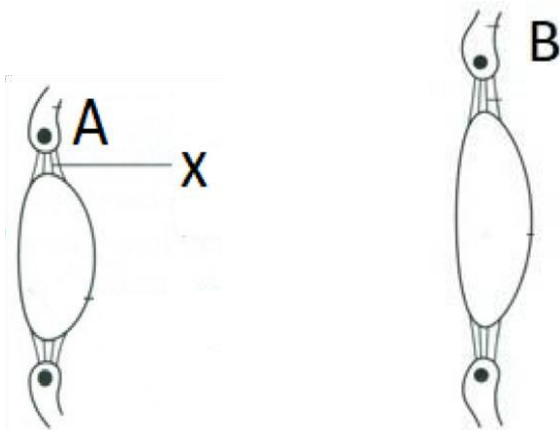
Caring for the Skin

- Wash the skin regularly to prevent accumulation of dirt and bad odour on the skin.

- Wear gloves when handling chemicals that can harm the skin.
- Protect your skin from direct radiation by wearing protective clothings
- Take food rich in vitamin C. Vitamin C keeps the skin soft and pliable.
- Dry your skin properly after washing to prevent fungal diseases.
- Examine the skin regularly for diseases. Seek medical help if you see any signs of disease.

Revision Exercise

- What is a sense organ?
 - Name sense organs that are responsible for detecting the following stimuli
 - touch - smell
 - sound
 - light
- The figure below shows different shapes of the lens of the eye viewing a distant and near object



- Name the part labelled X
- Which of the lens shows the eye

TOPIC 6: LOCOMOTION

- Locomotion is the movement of the whole organism from one place to another.

Reasons for Locomotion

- To run away from danger
- To search for mates
- To search for food

SKELETON

- A skeleton is a supportive framework in the body of an organism.

Types of Skeleton

a. Endoskeleton

- This is a type of skeleton located inside the body of an organism.
- The hard material is inside the soft tissue.
- The skeleton permits growth but does not offer maximum protection to soft tissues.

b. Exoskeletons

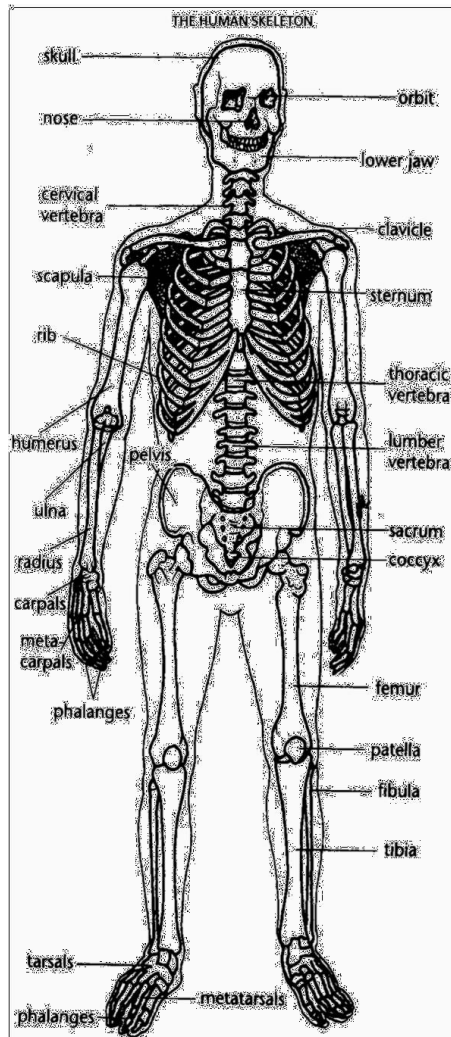
- It is a type of skeleton located on the outside of the body of an organism.
- The soft tissues are inside the hard material. It offers maximum protection to soft tissues.
- However, it hinders growth in organisms.
- Growth is possible after moulting (shedding of the exoskeleton).

c. Hydrostatic Skeleton

- A skeleton provided by a fluid enclosed in a cavity in the body of an organism.
- The fluid is always under high pressure.

THE HUMAN SKELETON

Main Parts of the Human Skeleton

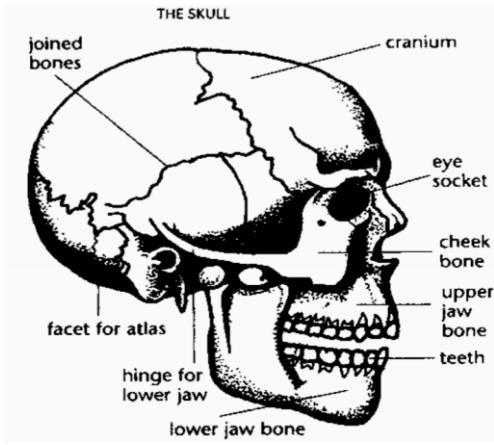
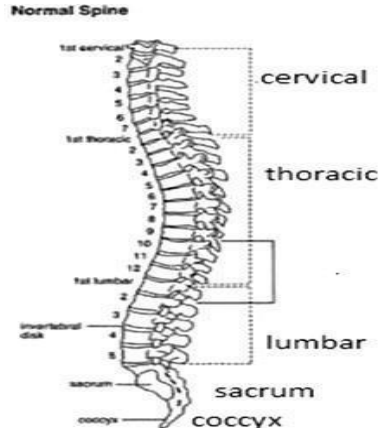
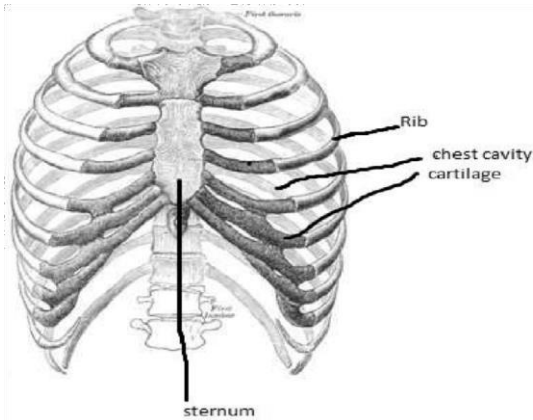


The human skeleton is divided into two main parts which are:

- i. The axial skeleton
- ii. The appendicular skeleton

The Axial Skeleton

- The word axial comes from the word axis.
- An axis refers to the central line of an organism or organ.
- The axial skeleton is made up of the skull, vertebral column and the rib cage.

Part	Diagram	Function
Skull		It protects the brain, nasal organs, eyes, middle and inner ears
vertebral column		It protects the spinal cord It supports the head It offers points of attachment to the pelvis and rib cage
Rib cage		It protects the lungs, heart, liver and other internal organs

The Appendicular Skeleton

- The word appendicular comes from the word append which means join to.
- Appendicular skeleton refers to part of the endoskeleton that joins the axial skeleton.
- The appendicular skeleton consists of the pectoral girdle, pelvic girdle and paired limbs.
- Fore limbs are attached to the pectoral girdles. Hind limbs are attached to the pelvic girdles.

Functions of Parts of the Appendicular Skeleton

a. The Pectoral Girdle -

It gives attachment to forelimbs.

- It also absorbs stress from forelimbs

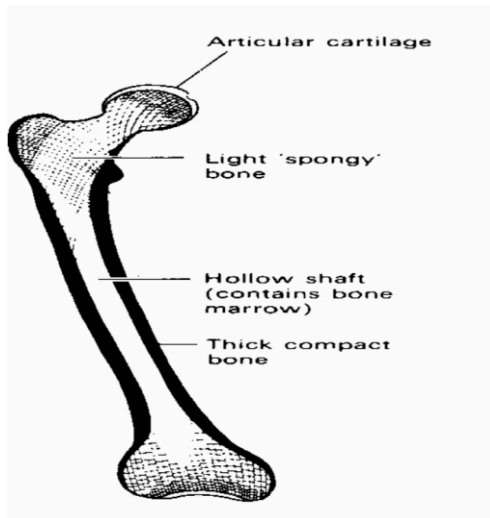
Pelvic Girdle

- It gives attachment to hind limbs
- It also absorbs stress from the hind ***Paired limbs (legs and arms)***

- For locomotion and movement
- Holding and grasping objects
- **Functions of Bones**

- Support:** the skeleton holds the rest of the body in position and helps in maintaining its shape.
- Protection:** the skeleton protects delicate organs
 - For example the spine protects the spinal cord and the skull protects the brain and eyes.
- Movement:** bones meet at joints which form levers.
 - Muscles pull on these levers to move the body
- Formation of blood cells:** blood cells are formed in the bone marrow.
 - Red blood cells are made in red bone marrow in short and long bones.
 - White blood cells are made in yellow bone marrow in shafts of long bones.
- Storage** bones also store minerals such as calcium and phosphorus

BoneStructure



- The bone consists of living bone cells called **osteoblasts**.
- The bone is made of protein called **collagen** and minerals (calcium and phosphorous).
- Magnesium and fluoride are also present but in small quantities.
- The bones have blood vessels that provide nutrients and oxygen to bone tissues and carry metabolic wastes from bones to excretory organs.
- The bones also have nerves.
- There are two types of bones, that is, compact bone and spongy bone.

Compact Bone

- It is very dense, heavy and strong.
- It is mainly found in shafts of long bones

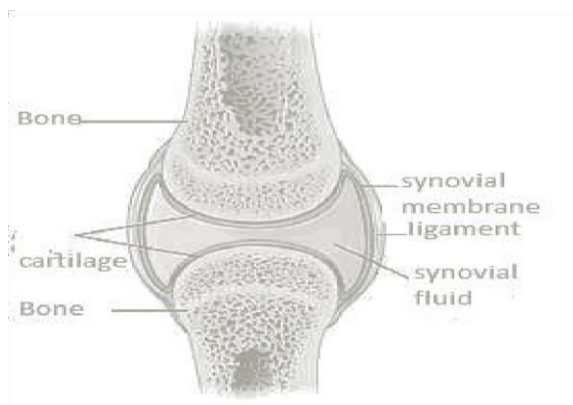
Spongy Bone

- The spongy bone has an open structure hence it is light.
- It is found where there are large masses of bone such as the head of femur.

Parts of Bone and Functions

Part	Function
Cartilage	Acts as shock absorber
Spongy bone	• Makes the bone light because it is very porous • It is also a site of blood cell formation because it has bone marrow
Compact bone	It is rigid and strong for support
Hollow shaft of the bone	It reduces the weight of the bone without reducing its strength
Yellow bone marrow (in long shaft of the bone)	Manufactures white blood cells
Red bone marrow (in spongy bone)	Manufactures red blood cells

NB: Long bones contain yellow bone marrow where white blood cells are made. The short bones contain red bone marrow and manufacture red blood cells.



JOINTS

- A joint is a place where two or more bones meet

Parts of a Joint and their Functions

Part of Joint	Function
Capsule	Binds the whole joint and keeps it intact
Synovial membrane	Secretes the synovial fluid
Synovial fluid	Lubricates the joint
Cartilage	Acts as shock absorber at a joint
Ligaments	Hold the bones at a joint together firmly

Types of Joints

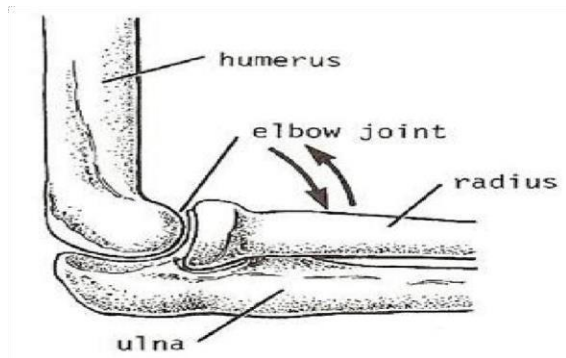
There are two main types of joints:

- **Movable or synovial** joints and **immovable joints**

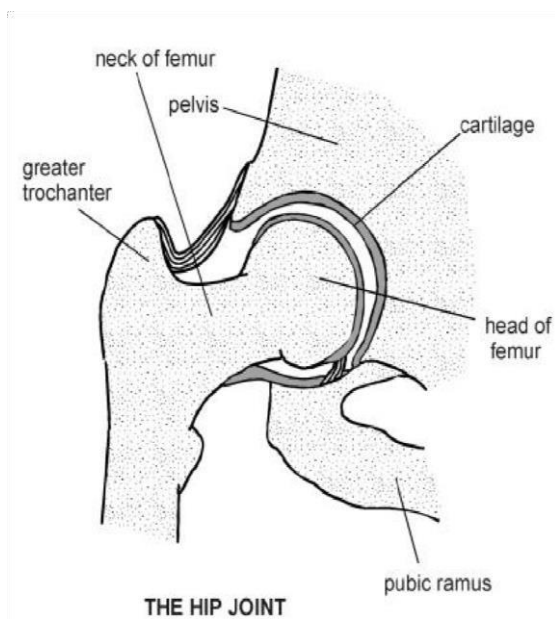
Movable Joints

- These are joints that allow free movement of bones to occur. Examples of movable joints are, hinge joint, ball and socket joint, gliding joint and peg and socket joint.

a. Hinge joint

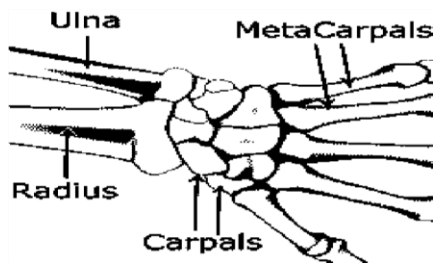


- A hinge joint allows movement in one plane only.
- Examples of hinge joints are elbow and knee joints.
- It is able to bear heavy loads.



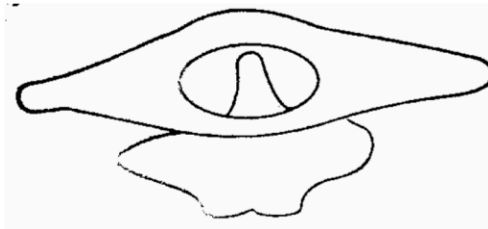
- A rounded head of one bone fits into a cup-shaped socket of another.
- It allows movement in all planes.
- It is unable to bear heavy loads.
- Examples of ball and socket joints are the shoulder and hip joints.

b. Gliding joints



- These allow movement by sliding over each other such as between bones of the vertebrae, wrist and ankles.

c. Peg and Socket (Pivot Joint)



- It is found in the vertebral column in the neck(the axis).
- It enables the head to rotate.

Immovable or Fixed Joints

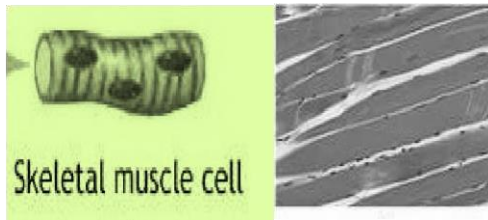
- These are joints that do not allow movement of the bones to occur.
- They are called sutures especially cranial joints.
- Examples of fixed joints are the cranial (of the skull) and pelvic joint.

Muscles

- Muscles form most of the soft tissues of the body.
- They work by contracting
- They are made of thousands of small fibres.

Types of Muscles

a. Skeletal or Voluntary Muscles



- These are muscles that are associated with the skeleton and are under control of a person's will.

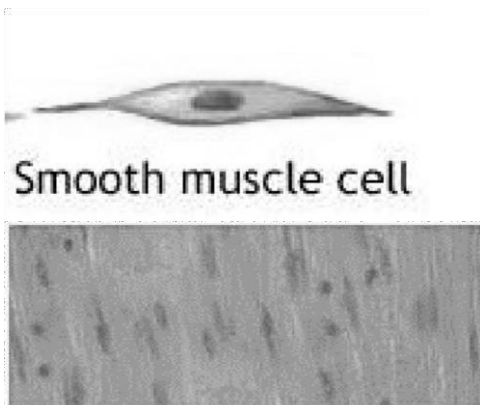
Characteristics of Skeletal or Voluntary Muscles

- They are under control of a person's will
- They are attached to the skeleton
- They can contract rapidly and tire easily
- They are striated or striped
- They are involved in locomotion
- They can remain contracted for a long time.

Functions of Skeletal Muscles

- They are involved in locomotion
- They help in holding, lifting and grasping things
- They also help in talking, chewing, etc.

b. Smooth or Involuntary Muscles



- These are muscles that occur mostly in hollow internal organs such as kidneys, blood vessels, intestines and bladder

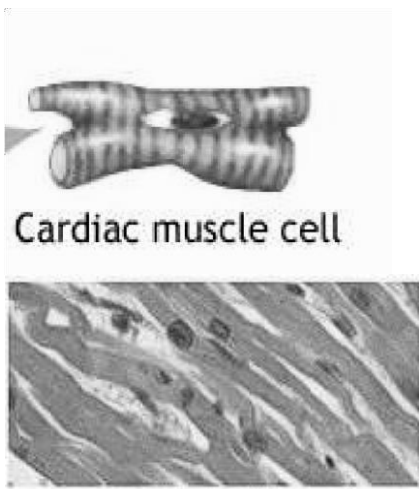
Characteristics of Involuntary Muscles

- They contract and tire slowly
- They are not striated or striped
- They are under control of the autonomic nervous system

Functions of Smooth or Involuntary Muscles

- They aid in digestion by contracting and relaxing to move food in the gut.
- They regulate movement of substances in tube as such anal sphincter muscles.

c. Cardiac or Heart Muscles



- These are muscles of the heart.

Characteristics of Cardiac Muscles

- They are striated
- They contract and relax without tiring
- They are under control of the autonomic nervous system
- They can contract even after the heart is removed from the body.

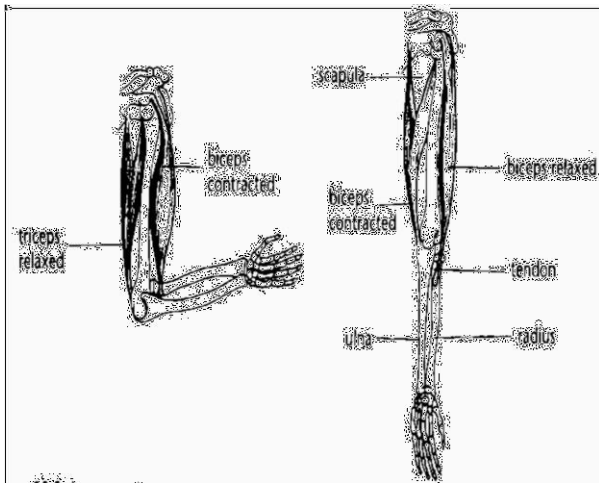
- Function of Cardiac Muscles

- They pump blood in the body.

Antagonistic Muscles

- These are pairs of skeletal muscles.
- When one muscle relaxes the other one relaxes.

- One example of antagonistic muscles is biceps and triceps on the inside and backside of the bone of the upper arm the humerus.



- Muscles that contract and bend limbs are called flexor muscles.
- The muscles that contract and extend limbs are called extensor limbs.

Stretching the arm

- The biceps relax.
- The triceps contract and pull at the bones of the lower arm (radius) to stretch the arm.

Bending the arm

- The triceps relax.
- The biceps contract and pull at the bones of the lower arm causing the arm to bend.

INJURIES TO BONES AND JOINTS

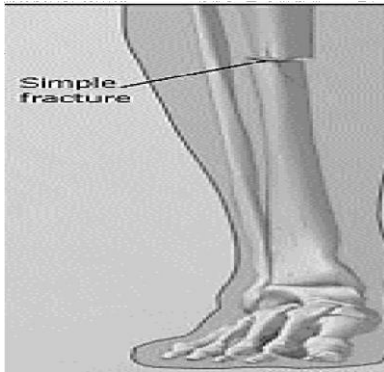
Fractures

- A fracture is a broken or cracked bone.
- Fractures occur when a lot of stress is applied or exerted on the bones.

Types of fractures

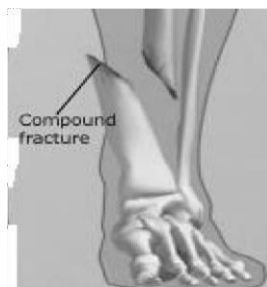
- a. Simple or closed fractures
- b. Compound or open fractures

Simple Fractures



- In an open fracture the bone remain beneath and does not protrude or pierce through the skin.

Compound fractures



In a compound fracture the broken bone protrudes or pierces through the skin.

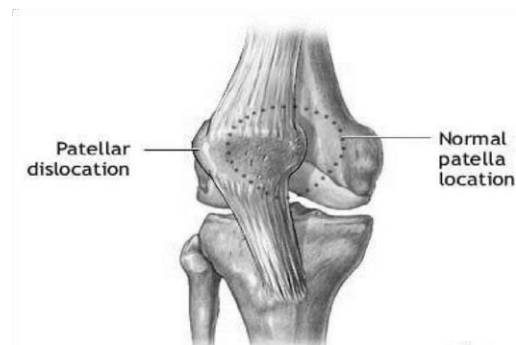
Signs of a Fracture

- A lot of pain at the site of the fracture.
- swelling and tenderness around the injury
- Bruising at the site of fracture
- Deformity-unusual positioning of the limb in question.
- Punctured or cut skin at the fracture in case of compound fracture.
-

First Aid for Fractures

- The first step is to call for an emergency as soon as the tragedy occurs. Do not try to move the patient if you suspect a neck or back or head injury to avoid making the injury worse.
- Never try to push the bone back because it can make the injury worse
- In case of bleeding, apply direct pressure, but never try to elevate the affected part for treatment.
- Once the bleeding stops cover the part with a clean dry cloth or dressing
- If the patient is in state of shock avoid giving them anything to eat or drink.
- If the wound is open, apply an icepack and cover it with a cloth. **The process of Immobilisation**
- in order to immobilize the fractured bone, check the extent of the fracture
- place a padded splint below the broken bone and wrap the splint in a cloth or towel for padding
- Tie the splint to the limb using ties, belts, pieces of cloth or bandages. Make sure that you do not bind it directly over the broken part.

DISLOCATION



- dislocation is a displacement of a bone at a joint.
- A bone moves out of position of the joint.
-

A dislocation usually occurs at the shoulder or knee joints.

First aid for Dislocations

- Carefully and skillfully push the dislocated bone back into position. This is done by a qualified first aider.
- Tie a sling along the affected area to support the weight away from the joint.

SPRAINS

- A sprain is an injury to ligaments and other tissues at a joint.
- It occurs when the joint is suddenly twisted or pulled.
- The twisting and pulling causes tear and injury to ligaments.

Symptoms of Sprains

- pain in the injured area
- swelling in the injured area
- difficulty in using or moving the injured area
- bruising in the injured area
- limping if the injury occurs to hip, ankle or knee.

First aid for Sprains

- Ice or cold water can be applied to reduce pain, swelling or bleeding
- Apply a supporting bandage at the joint and giving it enough rest so that the damaged tissue can heal.

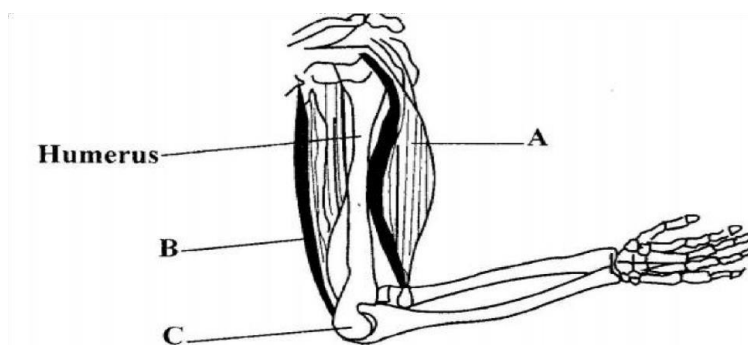
STRAINS

- A strain is a stretch or a tear on a muscle or tendon.
- A tendon is tissue that attaches muscle to the bone.
- Some strains heal by themselves while some require surgery.

NB: The first aid for Strains is similar to that of sprains.

- The first aid for sprains and strains can be summarized as **RICE** which stands for:
 - a. **Rest:** injured parts must be rested so that they heal faster
 - b. **Immobilise:** Stabilise the patient so that the injured part is not moved to avoid worsening the injury.

- c. **Cold:** apply cold compress to the injured area for 20-30 minutes for every two to three hours during the first two days.
 - d. **Elevation:** elevate the injured areas to limit internal bleeding and swelling as it reduces blood flow to the elevated parts.
1.
 - a. What is a skeleton?
 - b. Describe the three types of skeleton
 - c. Mention the three types of muscles
 2. The diagram below shows part of the human skeleton and associated muscles



- a. Name the type of joint formed at C
 - b. Describe how the lower arm is raised
 - c. What is a sprain?
 - d. State any two ways in which swelling around the sprain can be avoided
3.
 - a. What are muscles of the heart called?
 - b. What is a fracture?
 - c. Mention any two types of fractures
 - d. Why do we apply a cold compress on a site of fracture?
 5.
 - a. What is a dislocation?
 - b. Describe the first aid for a dislocation
 6. Give one function for each of the following parts of the skeleton
 - a. Skull
 - b. Rib cage
 - c. Spine
 - d. Eyesocket
 7. Describe any five functions of bones in an essay form

TOPIC 7: DIARRHOEAL DISEASES

Meaning of Diarrhoea

- A condition where by an individual passes out loose watery stools due to infection of the alimentary canal.

Examples of Diarrhoeal Diseases

1. Cholera
2. Dysentery
3. Typhoid

CHOLERA

- This is an acute (severe) infection of the intestinal tract.
- In serious cases a patient dies within 24 hours of catching the disease.

Cause of Cholera

- Cholera is caused by a bacterium called *Vibrio cholerae*.
- Once inside an individual, the bacterium has an incubation period of one to six days.
- Incubation period is the period between infection and appearance of symptoms of a disease.
- The bacterium produces toxins (poisonous substances) called enterotoxins that cause the symptoms of cholera.

Mode of Cholera transmission

- Faecal contamination
- Food contamination
- Vectors like flies from human faeces to food.
- Through drinking contaminated water

Signs and Symptoms of Cholera

- Severe diarrhoea due to toxins produced by bacteria
- Abdominal pains
- Watery stools that look like rice water
- Severe vomiting

- Dehydration due to loss of water in the body

Ways of Preventing Cholera

- Correct disposal of faeces
- Wash hands after using the toilet
- Wash hands before eating
- Treating water before drinking such as boiling or chlorination.
- Education awareness campaigns on hygiene
- Covering food to prevent contact with flies
- Wash raw food with clean water before eating

Treatment of Cholera

- Use of antibiotics
- Administering of ORS to the patient

DYSENTERY

- This is an intestinal infection that can lead to severe diarrhoea with mucus or blood in the faeces.

Causes of Dysentery

There are two main causes of dysentery:

- A bacterium called *shigella*. (causes bacillary dysentery)
- An amoeba called *Entamoeba histolytica*. (causes amoebic dysentery)
- The amoeba may begin feeding on the bacteria found in the intestines.
- It may later destroy the intestinal epithelial walls and blood capillaries leading to bleeding and formation of ulcers.
- The parasite starts feeding on red blood cells.
- It may later spread to the liver causing liver problems.

Mode of Dysentery Transmission

- Through eating contaminated food
- By houseflies
- Through drinking contaminated water

Signs and Symptoms of Dysentery

- Abdominal pain

- Bloody diarrhoea
- Flatulence
- Nausea with or without vomiting
- Anaemia in chronic cases of the disease

Prevention of Dysentery

- Drink treated
 - Wash hands with antibacterial soap after toilet, changing diapers and before preparing food.
- Eating properly cooked food
- Proper waste disposal
- Control flies

Treatment

- Use drugs such as **metronidazole** to treat amoebic dysentery
- Use antibiotics to treat bacillary dysentery
- Administer oral rehydration solution

TYPHOID

It is caused by bacteria called **salmonella typhi**. It affects the alimentary canal leading to ulcers and in extreme situations it breaks down the intestinal walls leading to death. The bacterium has an incubation period of two weeks.

Mode of Transmission

- Contaminated food and water
- Contaminated objects
- Vectors-flies or carriers (a carrier is an individual who can transmit a disease but is not showing its signs and symptoms)

Signs and Symptoms

- Mild fever
- Slight abdominal pain

- Ulceration and rupture of the intestine
- Diarrhoea
- Constipation

Prevention of Typhoid

- Vaccination
- Proper disposal of faeces
- Treatment of water
- Thorough cooking of food to kill bacteria
- Wash fruits and vegetables before eating them
- Isolation of patient to prevent spread of the disease through contact

Treatment of Typhoid

- Use of antibiotics

Ways of Preventing and Controlling Diarrhoeal Diseases

a. Water treatment:

- Water treatment kills germs such as bacteria and flukes hence preventing spread of diseases.
- There are many methods of water treatment such as filtration, boiling and using chemicals such as chlorination.

b. Proper disposal of human and domestic wastes:

- This includes using latrines, toilets and sewage treatment and rubbish pits.
- This helps in prevention of diseases such as cholera, typhoid and dysentery.

c. Personal hygiene: this

- Involves observing personal cleanliness such as washing hands before eating food and after using toilet.
- Personal hygiene reduces risk of diarrhoeal infection.

d. Vector control:

- Control of vectors such as flies and cockroaches prevents the spread of diarrhoeal diseases.

e. Food hygiene:

- Wash fruits and vegetables with clean running water, covering food, do not eat food sold in open places and cooking food properly.

- This reduces chances of diarrhoeal infection.

f. **Public awareness campaigns:**

- This enables people to be aware of various means of preventing diarrhoeal diseases.

Home-Made Treatment of Diarrhea

- A person loses a lot of water with diarrhoea.
- The body cells do not get enough water which can lead to death.
- To overcome dehydration, a person with diarrhoea is a solution called oral rehydration solution (ORS).
- The solution enhances faster absorption of water into the body.
- ORS is made by dissolving salts and sugar into water.
- It helps the body to replace fluids lost during diarrhoea.

How to Prepare Home-made ORS Materials

- Clean water
- Sugar
- Salt

Procedure

1. Put one litre of water in a jug
2. Add 10 teaspoons of sugar
3. Add 1 teaspoon of salt
4. Stir the mixture till the salt and sugar dissolve

The solution should be given regularly to the patient.

Revision Exercise

1. Name the causative agents for the following diseases
 - a. Cholera
 - b. Dysentery
 - c. Typhoid
2.
 - a. Describe any **five** ways in which diarrhoeal diseases can be transmitted
 - b. Describe any **five** ways in which diarrhoeal diseases can be prevented in our communities
 - c. Describe how you can prepare an oral rehydration solution
3. State any two signs and symptoms of the following diseases
 - a. Cholera
 - b. Dysentery

TOPIC 8: SEXUALLY TRANSMITTED INFECTIONS (STIs)

- A sexually transmitted infection is an infection that can be transferred from one person to another through sexual contact.

Examples of Sexually Transmitted Infections

- HIV & AIDS, Syphilis, gonorrhea, Chancroid, Candidiasis, genital warts and genital herpes.

GONORRHEA (chinsono)

- Gonorrhea is caused by a bacterium called *Neisseria gonorrhea*.
- The bacteria attacks the urethra in males and vagina in females.

Mode of Gonorrhea Transmission

- Sexual intercourse with infected persons
- At birth for newborn babies if the mother is infected

Signs and Symptoms of Gonorrhea

In Females

- Pain in the lower abdomen
- Menstrual problems
- Discharge of pus from the vagina

In Males

- Yellowish discharge from the urethra
- Pain while passing out urine

Effects of Gonorrhea Infections

- Blockage of sperm ducts or oviduct which leads to sterility
- Destruction of blood cells hence causing anaemia
- Destruction of the liver cells causing jaundice (yellowing of skin and eyes)

Control and Prevention of Gonorrhea

- Abstaining from sex before marriage
- Early diagnosis and treatment using antibiotics

- Engaging in safe sex through the use of condoms
- Health education to the community to control the spread

SYPHILIS (chindoko)

- Syphilis is caused by a bacterium called *treponema pallidum*.
- Like gonorrhea, it also affects the urethra and the vagina and is more serious than gonorrhea.

Mode of Syphilis Transmission

- Sexual intercourse with infected person
- At birth for newborn babies by infected mother

Signs and Symptoms of Syphilis Syphilis occurs in three phases **First Phase**

- Painful sore appear on the cervix or tip of the penis (chancre) **Second Phase**

- Rashes appear on the skin
- Falling of hair
- Mild fever
- Enlarged lymph nodes

The patient is highly infectious at this stage.

Third Phase

- The infection reaches the nervous system and the heart. It destroys them or causes paralysis, blindness or madness. Death may eventually occur. **Effects of Syphilis Infection**

- Heart diseases
- Blindness or insanity
- Paralysis
- Death

Control and Prevention of Syphilis

- Engaging in safe sex through the use of condoms
- Abstain from sex before marriage
- Health education to create awareness in the community
- Treat using antibiotics such as penicillin.

CANDIDIASIS

- Candidiasis is caused by a fungus called *Candida albicans*.
- Candidiasis are of two types:

Oropharyngeal candidiasis- This occurs when the fungus attacks the mouth. It is also known as oral thrush

- i. candidiasis-this occurs when the fungi attacks the vagina

Mode of Transmission of Candidiasis

- Through sexual intercourse with infected person
 - Oropharyngeal candidiasis is transmitted through kissing.
 - From mother to child during birth
- ### **Signs and Symptoms of Candidiasis**
- White patches in the infected area
 - Red inflamed skin under the patches
 - Severe irritation of the vagina
 - Vaginal discharge

Control and prevention of Candidiasis

- Abstaining from sex before marriage
- Use drugs such as clotrimazole
- Practice safe sex

GENITAL WARTS

Genital warts is caused by *Human Papilloma Virus* (HPV). The infection is very common among teens and young adults. The virus that causes genital warts is also linked to cervical cancer.

Mode of Transmission

- Sexual intercourse with an infected person.

Signs and Symptoms of Genital Warts

- Bump-like growths on the genitals and anus
- Itching or discomfort around the genitals
- Bleeding may occur

Prevention and Control

- Removal of genital warts using liquid nitrogen
- Abstaining from sex before marriage

HEPATITIS B

The disease is caused by the hepatitis virus.

Mode of Transmission of Hepatitis

- Through sexual intercourse with an infected person
- Through blood transfusion with infected blood

Signs and Symptoms of Hepatitis B

- Production of dark coloured urine
- Jaundice
- Inflamed liver which swells and stops functioning
- Pain in the stomach
- Muscle ache, fatigue, nausea, vomiting and diarrhoea.

Prevention and Control of Hepatitis B

- Abstaining from sex before marriage
- Engaging in protected sex by use of condoms
- Treatment using drugs

CHANCROID (Mabomu)

- This is a disease that is caused by bacteria.
- It is a disease that can be treated and cured with no long term effects.

Mode of Chancroid Transmission

- Sexual intercourse with infected person

Signs and Symptoms

- Painful ulcers on external genital organs
- Swelling in groin

In Females

- It causes abnormal discharge
- Painful urination (dysuria)
- Painful sexual intercourse (dyspareunia)

- Painless sores on the cervix
- Several sores around the vagina and rectum.

In Males

- 1-4 sores may develop on the penis
- Bubo appears in half of the male patients
- The foreskin may swell

Control and Prevention of Cancroids

- Abstain from sex before marriage
- Engage in safe sex
- Treat using antibiotics

GENITAL HERPES

- The disease is caused by a Herpes virus.

Mode of Transmission

- Through sexual intercourse
- At birth by an infected mother to the newborn baby

Signs and Symptoms

- Sores on the lips
- Painful blisters on the genitals
- Genital ulcers which may heal and recur

Control and Prevention

- Abstaining from sex before marriage
- Engage in safe sex by using condoms

CHLAMYDIA

- The disease is caused by a bacterium called chlamydia trachomatis.
- The disease is similar to gonorrhea in many ways and it is difficult to differentiate them.

Mode of Transmission

- Through sexual intercourse with infected person

Signs and Symptoms

- Inflammation of the pelvis
- Abnormal discharge from genitals
- Pain and discomfort especially during urinating
- Bleeding especially after sex
- Infertility as it affects women's fallopian tubes

Control and Prevention

- Abstaining from sex before marriage
- Treatment using antibiotics such as penicillin
- Using condoms

AIDS Meaning of AIDS

A: Acquired-means passed from one person to another

I: Immune-means the body's defence or protection against infections

D: Deficiency-means weakening of the body's defence against diseases

S: Syndrome-a group of diseases that come when the immunity is weak

AIDS is caused by Human Immunodeficiency Virus (HIV). The virus destroys the body's immune system.

Mode of Transmission of AIDS

- Through sexual intercourse with an infected person
- From infected mother to child during birth
- Sharing of piercing objects such as needles, syringes and razor blades
- Blood transfusion with infected blood
- Through close contact between infected and non-infected person through cuts or open wounds

Signs and Symptoms of AIDS

- Sudden loss of weight
- Lymphoma (cancer of the lymphatic system)
- Chronic diarrhoea for more than a month
- Persistent cough
- Shingles
- Fever

- Swollen lymph glands or lymph nodes
- Loss of appetite

Prevention of the Disease

- Abstinence from sex before marriage
- Practicing protected sex
- Being faithful to one partner
- Screening blood for HIV before bloodtransfusion
- Avoid sharing piercing objects such asblades, needles and syringes.

Treatment

- Use antiretroviral drugs (ARVs) whichslow down the spread of the virus in thebody. AIDS has no cure.

Some misconceptions about HIV & AIDS

- AIDS is a result of witchcraft
- HIV can be spread by mosquito bites
- HIV can be spread by sharing utensils
- HIV can be spread by kissing, hugging
- HIV can be cured traditionally
- AIDS is genocide to wipe minorityraces in USA
- People under ARVs cannot spread HIV

1. On Individuals

- Loss of body weight
- Madness in case of lack of treatment such as syphilis
- Sterility for example due gonorrhea
- Blindness in case of syphilis in babies
- It can lead to death such as AIDS
- Low productivity because the personsmay be weak to work.

2. On the Family

- Loss of income to the family when theinfected person is too weak to work
- In case of death, children are leftorphans and partners are widowed
- Burden to relatives in terms of care,medication, widows, etc.

3. On the Nation

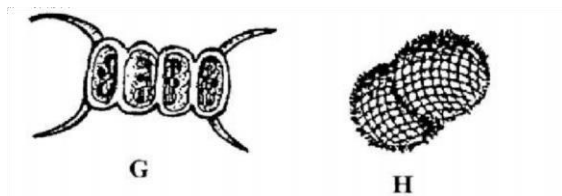
- High cost of medical care because the nation spends a lot of money on providing health care.
- Low productivity of the nation because people have reduced productivity.
- Loss of manpower to the nation due to death

Ways of Caring for HIV and AIDS Patients

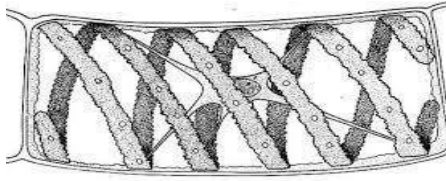
- Physical care such as washing their clothes, feeding them, bathing them and taking them to hospital.
- Providing emotional and spiritual support such as praying with them, encouraging them to live positively,
Accepting them in communities and at work.
- Providing them with nutritious food with balanced diet
- Ensuring that they observe hygiene such as taking a bath every day, washing hands after visiting the toilet
- Provision of medical care such as taking anti-retroviral drugs

Revision Exercise

1. a. List down five main groups of microorganisms
b. Mention any two negative effects of microorganisms
c. Give the habitat for algae
2. The diagram below shows microorganisms



- a. To which group of microorganisms do G and H belong?
 - b. Give any two observable differences between G and H
 - c. State any one importance of G in its habitat
3. The diagram below shows a microorganism



- a. Identify the group to which the microorganism belongs
 - b. Mention any two importance of the microorganism to its habitat
 - c. On the diagram label the nucleus, cytoplasm, strand and cell wall
4. a. Describe any five importance of microorganisms